



# Maintenance Schedule

Diesel engine

V4000R41x

V4000R42

Application group 2A

M050640/08E



A Rolls-Royce  
solution

| Engine model | kW/cyl.     | Application group                      |
|--------------|-------------|----------------------------------------|
| 8V4000R41R   | 106 kW/cyl. | 2A; Continuous operation, unrestricted |
| 8V4000R41    | 125 kW/cyl. | 2A; Continuous operation, unrestricted |
| 8V4000R41L   | 138 kW/cyl. | 2A; Continuous operation, unrestricted |
| 12V4000R41R  | 115 kW/cyl. | 2A; Continuous operation, unrestricted |
| 12V4000R41   | 125 kW/cyl. | 2A; Continuous operation, unrestricted |
| 12V4000R41L  | 138 kW/cyl. | 2A; Continuous operation, unrestricted |
| 16V4000R41R  | 106 kW/cyl. | 2A; Continuous operation, unrestricted |
| 16V4000R41   | 125 kW/cyl. | 2A; Continuous operation, unrestricted |
| 16V4000R41L  | 138 kW/cyl. | 2A; Continuous operation, unrestricted |
| 20V4000R42   | 135 kW/cyl. | 2A; Continuous operation, unrestricted |

Table 1: Valid for

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# 1 Maintenance Schedule

## 1.1 Preface

This manual provides instructions for your product from the manufacturer MTU.

### Definition of MTU

MTU refers to Rolls-Royce Power Systems AG and MTU Friedrichshafen GmbH or an affiliated company pursuant to Section § 15 AktG (German Stock Corporation Act) or a controlled company (joint venture).

### Maintenance concept

Information about the latest maintenance schedule is available at: <http://www.mtu-solutions.com>.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability. The maintenance intervals as well as the required scopes of maintenance tasks are based on operational experience and therefore to be considered as recommendations. Additional maintenance work and/or changes to the maintenance intervals may be required in the case of difficult operational and ambient conditions.

Adherence to the specified maintenance intervals is essential to maintain product safety.

The intervals according to which the maintenance tasks have to be carried out are specified as operating hours and time limits. Whichever value is reached first shall apply.

The maintenance intervals are determined and verified by the load profile. Load profiles can be evaluated by the Service Partner of MTU. Reading out load profile data is recommended for the first time after between 500 and 1000 operating hours depending on the application concerned, or after changing the duty profile or operating area.

The individual maintenance intervals are assigned to the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.

QL2: Exchange of components and parts in case of repair (corrective only, not part of the maintenance schedule).

QL3: Maintenance work which requires partial disassembly of the product.

QL4: Maintenance work which requires complete disassembly of the product.

### Maintenance task tolerances

Maintenance tasks shall be completed within certain tolerances comprising a percentage and a maximum value, whereby the smaller value shall apply. Taking full advantage of such tolerances does not influence the scheduled timing of the ensuing maintenance task.

| Interval (hrs)/limit | Limit in %           | Limit max.                                                           |
|----------------------|----------------------|----------------------------------------------------------------------|
| Operating hours      | ± 10%                | ± 500 hrs                                                            |
| Years                | ± 10%                | ± 30 days                                                            |
| Example 1            | Task every 1000 hrs  | Between 900 hrs and 1100 hrs (10% of 1000 hrs)                       |
| Example 2            | Task every 30000 hrs | Between 29500 hrs and 30500 hrs (10% of 30000 hrs, but max. 500 hrs) |

### Additional notes on maintenance and preservation:

The relevant component manufacturer's instructions apply with respect to the maintenance of any components which do not appear in this maintenance schedule.

For the change intervals of fluids and lubricants, refer to the corresponding Fluids and Lubricants Specifications.

If the engine is to remain out of service for a longer time period, MTU recommends to carry out a monthly test run until engine operating temperature is reached. If the engine is scheduled to remain out of service for > 1 month, carry out engine preservation procedures in accordance with the Preservation and Represervation Specifications

The latest version of the specifications is available at <http://www.mtu-solutions.com>.

## 1.2 Allocation matrix application group 2A

### Note:

TBO identifiable via cylinder power (shown on engine ID plate or in 'Engine Data' in Business Portal, power/ number of cylinder) and, where appropriate, in contractually agreed load profile.

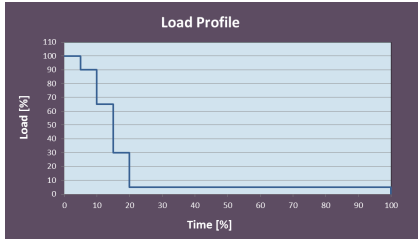
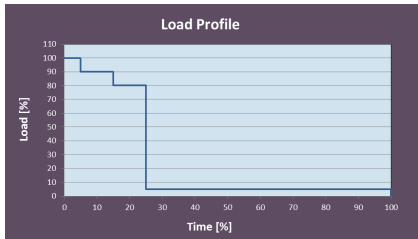
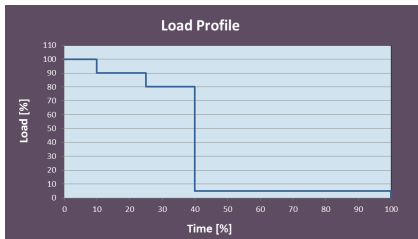
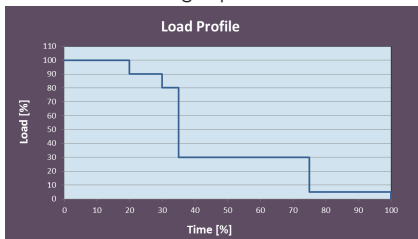
| Load profile             |        | 4000 R41x; R42 |                  |                       |                         |                         |   |                                                                                                            |
|--------------------------|--------|----------------|------------------|-----------------------|-------------------------|-------------------------|---|------------------------------------------------------------------------------------------------------------|
| Load %                   | Time % | Load factor %  | Load indicator % | From 0 to 106 kW/cyl. | From 107 to 125 kW/cyl. | From 126 to 138 kW/cyl. |   |                                                                                                            |
| 110 <sub>(110-100)</sub> | —      | From 0 to 18   | From 0 to 7      | 42000                 | 36000                   | 36000                   | — | <p>Switcher</p>        |
| 100 <sub>(100-90)</sub>  | 5      |                |                  |                       |                         |                         |   |                                                                                                            |
| 90 <sub>(90-80)</sub>    | 5      |                |                  |                       |                         |                         |   |                                                                                                            |
| 80 <sub>(80-65)</sub>    | —      |                |                  |                       |                         |                         |   |                                                                                                            |
| 65 <sub>(65-50)</sub>    | 5      |                |                  |                       |                         |                         |   |                                                                                                            |
| 50 <sub>(50-30)</sub>    | —      |                |                  |                       |                         |                         |   |                                                                                                            |
| 30 <sub>(30-0)</sub>     | 5      |                |                  |                       |                         |                         |   |                                                                                                            |
| 5 <sub>(idle)</sub>      | 80     |                |                  |                       |                         |                         |   |                                                                                                            |
| 110 <sub>(110-100)</sub> | —      | From 19 to 26  | From 8 to 10     | 42000                 | 36000                   | 30000                   | — | <p>Multi Purpose</p>  |
| 100 <sub>(100-90)</sub>  | 5      |                |                  |                       |                         |                         |   |                                                                                                            |
| 90 <sub>(90-80)</sub>    | 10     |                |                  |                       |                         |                         |   |                                                                                                            |
| 80 <sub>(80-65)</sub>    | 10     |                |                  |                       |                         |                         |   |                                                                                                            |
| 65 <sub>(65-50)</sub>    | —      |                |                  |                       |                         |                         |   |                                                                                                            |
| 50 <sub>(50-30)</sub>    | —      |                |                  |                       |                         |                         |   |                                                                                                            |
| 30 <sub>(30-0)</sub>     | —      |                |                  |                       |                         |                         |   |                                                                                                            |
| 5 <sub>(idle)</sub>      | 75     |                |                  |                       |                         |                         |   |                                                                                                            |
| 110 <sub>(110-100)</sub> | —      | From 27 to 38  | From 11 to 17    | 36000                 | 30000                   | 27000                   | — | <p>Standard</p>       |
| 100 <sub>(100-90)</sub>  | 10     |                |                  |                       |                         |                         |   |                                                                                                            |
| 90 <sub>(90-80)</sub>    | 15     |                |                  |                       |                         |                         |   |                                                                                                            |
| 80 <sub>(80-65)</sub>    | 15     |                |                  |                       |                         |                         |   |                                                                                                            |
| 65 <sub>(65-50)</sub>    | —      |                |                  |                       |                         |                         |   |                                                                                                            |
| 50 <sub>(50-30)</sub>    | —      |                |                  |                       |                         |                         |   |                                                                                                            |
| 30 <sub>(30-0)</sub>     | —      |                |                  |                       |                         |                         |   |                                                                                                            |
| 5 <sub>(idle)</sub>      | 60     |                |                  |                       |                         |                         |   |                                                                                                            |
| 110 <sub>(110-100)</sub> | —      | From 39 to 46  | From 18 to 25    | 30000                 | 27000                   | 24000                   | — | <p>High Speed</p>     |
| 100 <sub>(100-90)</sub>  | 20     |                |                  |                       |                         |                         |   |                                                                                                            |
| 90 <sub>(90-80)</sub>    | 10     |                |                  |                       |                         |                         |   |                                                                                                            |
| 80 <sub>(80-65)</sub>    | 5      |                |                  |                       |                         |                         |   |                                                                                                            |
| 65 <sub>(65-50)</sub>    | —      |                |                  |                       |                         |                         |   |                                                                                                            |
| 50 <sub>(50-30)</sub>    | —      |                |                  |                       |                         |                         |   |                                                                                                            |
| 30 <sub>(30-0)</sub>     | 40     |                |                  |                       |                         |                         |   |                                                                                                            |
| 5 <sub>(idle)</sub>      | 25     |                |                  |                       |                         |                         |   |                                                                                                            |

Table 2: Allocation matrix application group 2A

### 1.3 Noncyclical maintenance tasks

| No.                                                                                                                                                                                                             | Qualification | Task    | Maintenance tasks                                                                                                                                                                          | Reference task | Offset | Value | Unit | Option |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|--------|-------|------|--------|
| 0001                                                                                                                                                                                                            | QL1           | WM00094 | Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul. |                | 0      | 1000  | HRS  |        |
| Reference task: Name of the maintenance task the offset refers to<br>Offset: Interval of a maintenance task which depends on another interval<br>Value: Fixed value for binding execution of a maintenance task |               |         |                                                                                                                                                                                            |                |        |       |      |        |

## 1.4 Maintenance Schedule Matrix 42000 HRS

### 500 to 15000 Operating hours

| Task    | Limit | Operating hours [HRS] |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 500                   | 1000 | 1500 | 2000 | 2500 | 3000 | 3500 | 4000 | 4500 | 5000 | 5500 | 6000 | 6500 | 7000 | 7500 | 8000 | 8500 | 9000 | 9500 | 10000 | 10500 | 11000 | 11500 | 12000 | 12500 | 13000 | 13500 | 14000 | 14500 | 15000 |
| WM00285 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00286 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00287 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00288 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00289 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00290 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00291 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00881 | 1 MON |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00067 | 2 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00116 | 2 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00126 | 5 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00128 | 5 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00063 | 2 YR  | X                     | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00066 | 9 YR  | X                     | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00149 | 1 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00084 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00086 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00170 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00050 | 1 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00136 | 1 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00076 | 3 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00664 | 3 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00692 | 5 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00052 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00094 | 18 YR |                       | X    |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00961 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00962 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00119 | 3 YR  |                       |      |      |      |      |      |      |      |      |      |      | X    |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00110 | 9 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       | X     |       |       |       |       |       |       |       |       |       |
| WM00090 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       | X     |       |       |       |       |       |       |       |       |       |
| WM00098 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       | X     |       |       |       |       |       |       |       |       |       |
| WM00283 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |       |
| WM00976 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |       |
| WM00008 | 6 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00065 | 9 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00054 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00056 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00062 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00093 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00096 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00921 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |

DAY = Days

MON = Months

YR = Years

HRS = Hours

| Task    | Limit | Operating hours [HRS] |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
|---------|-------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|
|         |       | 500                   | 1000 | 1500 | 2000 | 2500 | 3000 | 3500 | 4000 | 4500 | 5000 | 5500 | 6000 | 6500 | 7000 | 7500 | 8000 | 8500 | 9000 | 9500 | 10000 | 10500 | 11000 | 11500 | 12000 | 12500 | 13000 | 13500 | 14000 | 14500 | 15000 |  |
| WM00970 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00971 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00973 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00974 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00981 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00006 | 6 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00011 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00072 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00087 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00091 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00980 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00009 | 15 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00001 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00002 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00045 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00061 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00070 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00088 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00092 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00095 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00099 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00100 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00103 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00104 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00108 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00111 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00127 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00137 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00139 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00353 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00439 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00925 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM00936 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |
| WM01030 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |  |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

## 15500 to 30000 Operating hours

| Task    | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 15500                 | 16000 | 16500 | 17000 | 17500 | 18000 | 18500 | 19000 | 19500 | 20000 | 20500 | 21000 | 21500 | 22000 | 22500 | 23000 | 23500 | 24000 | 24500 | 25000 | 25500 | 26000 | 26500 | 27000 | 27500 | 28000 | 28500 | 29000 | 29500 | 30000 |
| WM00285 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00286 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

| Task    | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 15500                 | 16000 | 16500 | 17000 | 17500 | 18000 | 18500 | 19000 | 19500 | 20000 | 20500 | 21000 | 21500 | 22000 | 22500 | 23000 | 23500 | 24000 | 24500 | 25000 | 25500 | 26000 | 26500 | 27000 | 27500 | 28000 | 28500 | 29000 | 29500 | 30000 |
| WM00287 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00288 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00289 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00290 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00291 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00881 | 1 MON |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00067 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00116 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00126 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00128 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00063 | 2 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00066 | 9 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00149 | 1 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00084 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00086 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00170 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00050 | 1 YR  |                       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       | X     |       |       |       | X     |       |       |       | X     |       |       |       |       | X     |       |
| WM00136 | 1 YR  |                       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       | X     |       |       |       | X     |       |       |       | X     |       |       |       |       | X     |       |
| WM00076 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       | X     |       |       |       | X     |       |       |       | X     |       |       |       |       | X     |       |
| WM00664 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       | X     |       |       |       | X     |       |       |       | X     |       |       |       |       | X     |       |
| WM00692 | 5 YR  |                       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       | X     |       |       |       | X     |       |       |       | X     |       |       |       |       | X     |       |
| WM00052 | 18 YR |                       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       | X     |       |       |       | X     |       |       |       | X     |       |       |       |       | X     |       |
| WM00094 | 18 YR |                       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       | X     |       |       |       | X     |       |       |       | X     |       |       |       |       | X     |       |
| WM00961 | 18 YR |                       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       | X     |       |       |       | X     |       |       |       | X     |       |       |       |       | X     |       |
| WM00962 | 18 YR |                       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       | X     |       |       |       | X     |       |       |       | X     |       |       |       |       | X     |       |
| WM00119 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       | X     |       |
| WM00110 | 9 YR  |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00090 | 18 YR |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00098 | 18 YR |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00283 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |
| WM00976 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |
| WM00008 | 6 YR  |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00065 | 9 YR  |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00054 | 18 YR |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00056 | 18 YR |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00062 | 18 YR |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00093 | 18 YR |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00096 | 18 YR |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00921 | 18 YR |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00970 | 18 YR |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00971 | 18 YR |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00973 | 18 YR |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00974 | 18 YR |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00981 | 18 YR |                       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00006 | 6 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

[illegible]

### 30500 to 42000 Operating hours

| Task         | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
|--------------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|--|--|--|--|
|              |       | 30500                 | 31000 | 31500 | 32000 | 32500 | 33000 | 33500 | 34000 | 34500 | 35000 | 35500 | 36000 | 36500 | 37000 | 37500 | 38000 | 38500 | 39000 | 39500 | 40000 | 40500 | 41000 | 41500 | 42000 |  |  |  |  |  |  |
| WM00285      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00286      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00287      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00288      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00289      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00290      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00291      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00881      | 1 MON |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| DAY = Days   |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| MON = Months |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| YR = Years   |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| HRS = Hours  |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |

| Task    | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |
|---------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|--|--|
|         |       | 30500                 | 31000 | 31500 | 32000 | 32500 | 33000 | 33500 | 34000 | 34500 | 35000 | 35500 | 36000 | 36500 | 37000 | 37500 | 38000 | 38500 | 39000 | 39500 | 40000 | 40500 | 41000 | 41500 | 42000 |  |  |  |  |
| WM00067 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |
| WM00116 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |
| WM00126 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |
| WM00128 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |
| WM00063 | 2 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |  |  |  |  |
| WM00066 | 9 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |  |  |  |  |
| WM00149 | 1 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |  |  |  |  |
| WM00084 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |  |  |  |  |
| WM00086 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |  |  |  |  |
| WM00170 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |  |  |  |  |
| WM00050 | 1 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |
| WM00136 | 1 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |
| WM00076 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |
| WM00664 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |
| WM00692 | 5 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |
| WM00052 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |
| WM00094 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |
| WM00961 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |
| WM00962 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |
| WM00119 | 3 YR  |                       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00110 | 9 YR  |                       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00090 | 18 YR |                       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00098 | 18 YR |                       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00283 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00976 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00008 | 6 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00065 | 9 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00054 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00056 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00062 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00093 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00096 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00921 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00970 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00971 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00973 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00974 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00981 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00006 | 6 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00011 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00072 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00087 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00091 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00980 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |
| WM00009 | 15 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

| Task    | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 30500                 | 31000 | 31500 | 32000 | 32500 | 33000 | 33500 | 34000 | 34500 | 35000 | 35500 | 36000 | 36500 | 37000 | 37500 | 38000 | 38500 | 39000 | 39500 | 40000 |
| WM00001 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00002 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00045 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00061 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00070 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00088 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00092 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00095 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00099 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00100 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00103 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00104 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00108 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00111 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00127 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00137 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00139 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00353 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00439 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00925 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00936 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM01030 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

## 1.5 Cyclical maintenance tasks 42000 HRS

| Qualification | Interval [HRS] | Limit | Item                                                               | Maintenance tasks                                                                                                                                                                          | Option | Task    |
|---------------|----------------|-------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| <b>Engine</b> |                |       |                                                                    |                                                                                                                                                                                            |        |         |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check engine oil level                                                                                                                                                                     |        | WM00285 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Visually inspect for leaks and general condition of engine                                                                                                                                 |        | WM00286 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check maintenance indicator of air filter                                                                                                                                                  | X      | WM00287 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check relief bore of coolant pump                                                                                                                                                          |        | WM00288 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | When engine is running, check for abnormal running noises, exhaust gas color and vibrations                                                                                                |        | WM00289 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Drain water and contaminants from fuel prefilter                                                                                                                                           | X      | WM00290 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check pointer of vacuum gauge of fuel prefilter                                                                                                                                            | X      | WM00291 |
| QL 1          | -              | 1 MON | BENDING / TORSIONALLY RESILIENT COUPLING AND STEEL-SPRING COUPLING | Check coupling (visual inspection), replace as necessary                                                                                                                                   | X      | WM00881 |
| QL 1          | -              | 2 YR  | LUBE OIL SYSTEM                                                    | Replace oil filter at every engine oil change, or when the time limit (years) is reached, at the latest                                                                                    | X      | WM00067 |
| QL 1          | -              | 2 YR  | COOLANT SYSTEM                                                     | Replace filter                                                                                                                                                                             |        | WM00116 |
| QL 1          | -              | 5 YR  | ENGINE MOUNTING                                                    | Measure height of rubber element                                                                                                                                                           |        | WM00126 |
| QL 1          | -              | 5 YR  | ENGINE MOUNTING                                                    | Measure buffer play, adjust if necessary                                                                                                                                                   |        | WM00128 |
| QL 1          | 500            | 2 YR  | LUBE OIL SYSTEM                                                    | Automatic centrifugal oil filter, check layer thickness of oil residue, clean, and replace paper element at every engine oil change at the latest                                          |        | WM00063 |
| QL 1          | 500            | 9 YR  | AUTOMATIC OIL FILTER                                               | Automatic oil filter, check and clean filtering screen                                                                                                                                     | X      | WM00066 |
| QL 1          | 1000           | 1 YR  | AIR SUPPLY SYSTEM                                                  | Clean air filter, empty dust collector box                                                                                                                                                 | X      | WM00149 |
| QL 1          | 1000           | 2 YR  | FUEL SYSTEM (LOW PRESSURE)                                         | Fuel prefilter, replace filter                                                                                                                                                             | X      | WM00084 |
| QL 1          | 1000           | 2 YR  | FUEL SYSTEM (LOW PRESSURE)                                         | Replace easy-change fuel filter                                                                                                                                                            |        | WM00086 |
| QL 1          | 1000           | 2 YR  | AUXILIARY POWER TAKE-OFF                                           | Check belt condition, replace if necessary. Adjust tension.                                                                                                                                | X      | WM00170 |
| QL 1          | 3000           | 1 YR  | ENGINE MOUNTING                                                    | Check general condition of rubber mounts (visual inspection)                                                                                                                               |        | WM00050 |
| QL 1          | 3000           | 1 YR  | ENGINE MOUNTING                                                    | Check securing screws for secure seating                                                                                                                                                   |        | WM00136 |
| QL 1          | 3000           | 3 YR  | AIR SUPPLY SYSTEM                                                  | Replace air filter                                                                                                                                                                         | X      | WM00076 |
| QL 1          | 3000           | 3 YR  | COOLANT SYSTEM                                                     | Replace valve cover                                                                                                                                                                        | X      | WM00664 |
| QL 1          | 3000           | 5 YR  | CRANKCASE BREATHER                                                 | Replace filter or filter element (HE200)                                                                                                                                                   | X      | WM00692 |
| QL 1          | 3000           | 18 YR | BATTERY-CHARGING GENERATOR                                         | Check condition of coupling                                                                                                                                                                | X      | WM00052 |
| QL 1          | 3000           | 18 YR | VALVE GEAR                                                         | Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul. |        | WM00094 |
| QL 1          | 3000           | 18 YR | GROUNDING SYSTEM                                                   | Replace carbon brushes at flywheel                                                                                                                                                         | X      | WM00961 |
| QL 1          | 3000           | 18 YR | GROUNDING SYSTEM                                                   | Replace carbon brushes and cylindrical roller bearings of grounding equipment at vibration damper                                                                                          | X      | WM00962 |
| QL 1          | 6000           | 3 YR  | CRANKCASE BREATHER                                                 | Replace filter or filter element                                                                                                                                                           | X      | WM00119 |
| QL 1          | 10500          | 9 YR  | RUNNING GEAR                                                       | Inspect cylinder liner with endoscope                                                                                                                                                      |        | WM00110 |
| QL 1          | 10500          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace fuel injection valve/injector                                                                                                                                                      |        | WM00090 |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

TIM-ID: 0000000000 - 000

| Qualification | Interval [HRS] | Limit | Item                                                               | Maintenance tasks                                                                                                          | Option | Task    |
|---------------|----------------|-------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------|---------|
| QL1           | 10500          | 18 YR | CYLINDER HEAD                                                      | Measure valve protrusion                                                                                                   |        | WM00098 |
| QL3           | 14000          | 18 YR | AUXILIARY POWER TAKE-OFF                                           | Replace auxiliary power take-off                                                                                           | X      | WM00283 |
| QL3           | 14000          | 18 YR | BATTERY-CHARGING GENERATOR                                         | Overhaul or replace battery-charging generator                                                                             | X      | WM00976 |
| QL3           | 21000          | 6 YR  | RUBBER SLEEVE                                                      | Replace rubber sleeve                                                                                                      |        | WM00008 |
| QL3           | 21000          | 9 YR  | AUTOMATIC OIL FILTER                                               | Automatic oil filter, remove and clean filter housing, replace filter element and bearing                                  | X      | WM00065 |
| QL3           | 21000          | 18 YR | COOLANT SYSTEM                                                     | Check LT thermostat housing and replace thermal actuator                                                                   |        | WM00054 |
| QL3           | 21000          | 18 YR | COOLANT SYSTEM                                                     | Check HT thermostat housing and replace thermal actuator                                                                   |        | WM00056 |
| QL3           | 21000          | 18 YR | LUBE OIL SYSTEM                                                    | Check centrifugal oil filter for wear                                                                                      |        | WM00062 |
| QL3           | 21000          | 18 YR | VALVE GEAR                                                         | Check rocker arm and valve bridge for wear. Visually inspect swing follower with endoscope through pushrod bore            |        | WM00093 |
| QL3           | 21000          | 18 YR | VALVE GEAR                                                         | Visually inspect camshaft running surfaces with endoscope through pushrod bore                                             |        | WM00096 |
| QL3           | 21000          | 18 YR | COOLANT SYSTEM                                                     | Replace sealing material for installation of coolant pump(s) and thermostat(s) (only LT and HT)                            |        | WM00921 |
| QL3           | 21000          | 18 YR | EXHAUST TURBOCHARGER                                               | Overhaul or replace exhaust turbocharger                                                                                   |        | WM00970 |
| QL3           | 21000          | 18 YR | STARTING SYSTEM                                                    | Overhaul or replace starter                                                                                                |        | WM00971 |
| QL3           | 21000          | 18 YR | COOLANT SYSTEM                                                     | Overhaul or replace coolant pump in LT circuit                                                                             |        | WM00973 |
| QL3           | 21000          | 18 YR | COOLANT SYSTEM                                                     | Overhaul or replace coolant pump in HT circuit                                                                             |        | WM00974 |
| QL3           | 21000          | 18 YR | LUBE OIL SYSTEM                                                    | Clean oil cooler and check for leaks or replace                                                                            |        | WM00981 |
| QL3           | 42000          | 6 YR  | HOSE                                                               | Replace hose line                                                                                                          |        | WM00006 |
| QL3           | 42000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace HP fuel sensor                                                                                                     |        | WM00011 |
| QL3           | 42000          | 18 YR | EXHAUST SYSTEM                                                     | Replace exhaust pipe bellows                                                                                               |        | WM00072 |
| QL3           | 42000          | 18 YR | FUEL SYSTEM (LOW PRESSURE)                                         | Replace pump                                                                                                               |        | WM00087 |
| QL3           | 42000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace HP pump                                                                                                            |        | WM00091 |
| QL3           | 42000          | 18 YR | CYLINDER HEAD                                                      | Overhaul or replace cylinder head                                                                                          |        | WM00980 |
| QL4           | -              | 15 YR | STARTING SYSTEM                                                    | Replace CAPOS                                                                                                              | X      | WM00009 |
| QL4           | 42000          | 18 YR | ENGINE, GENERAL                                                    | Replace elastomer parts and seals                                                                                          |        | WM00001 |
| QL4           | 42000          | 18 YR | ENGINE, GENERAL                                                    | Disassemble engine completely. Check engine components according to assembly instructions, replace or repair as necessary. |        | WM00002 |
| QL4           | 42000          | 18 YR | BENDING / TORSIONALLY RESILIENT COUPLING AND STEEL-SPRING COUPLING | Check coupling, replace if necessary                                                                                       | X      | WM00045 |
| QL4           | 42000          | 18 YR | OIL REPLENISHING UNIT                                              | Replace gear pump                                                                                                          | X      | WM00061 |
| QL4           | 42000          | 18 YR | LUBE OIL SYSTEM                                                    | Check pump, replace if necessary                                                                                           |        | WM00070 |
| QL4           | 42000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace pressure relief valve                                                                                              |        | WM00088 |
| QL4           | 42000          | 18 YR | VALVE GEAR                                                         | Replace swing follower and swing follower shaft                                                                            |        | WM00092 |
| QL4           | 42000          | 18 YR | VALVE GEAR                                                         | Replace thrust bearing flange                                                                                              |        | WM00095 |
| QL4           | 42000          | 18 YR | RUNNING GEAR                                                       | Replace conrod bearings                                                                                                    |        | WM00099 |
| QL4           | 42000          | 18 YR | RUNNING GEAR                                                       | Replace rings                                                                                                              |        | WM00100 |
| QL4           | 42000          | 18 YR | RUNNING GEAR                                                       | Replace crankshaft bearings                                                                                                |        | WM00103 |
| QL4           | 42000          | 18 YR | GEAR TRAIN                                                         | Check tooth flanks in gear train for wear (visual inspection), replace bushing                                             |        | WM00104 |
| QL4           | 42000          | 18 YR | RUNNING GEAR                                                       | Replace cylinder liners                                                                                                    |        | WM00108 |
| QL4           | 42000          | 18 YR | CAMSHAFT                                                           | Replace camshaft bearings                                                                                                  |        | WM00111 |
| QL4           | 42000          | 18 YR | ENGINE MOUNTING                                                    | Replace rubber element of rubber mounts                                                                                    |        | WM00127 |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

| Qualification                                           | Interval<br>[HRS] | Limit | Item                                     | Maintenance tasks                                                | Option | Task    |
|---------------------------------------------------------|-------------------|-------|------------------------------------------|------------------------------------------------------------------|--------|---------|
| QL4                                                     | 42000             | 18 YR | SENSOR TECHNOLOGY AND WIRING HARNESS     | Replace wiring harness                                           |        | WM00137 |
| QL4                                                     | 42000             | 18 YR | GROUNDING SYSTEM                         | Check flywheel (insulated), overhaul if necessary                | X      | WM00139 |
| QL4                                                     | 42000             | 18 YR | RUNNING GEAR                             | Replace conrod                                                   |        | WM00353 |
| QL4                                                     | 42000             | 18 YR | RUNNING GEAR                             | Replace vibration damper                                         |        | WM00439 |
| QL4                                                     | 42000             | 18 YR | BENDING / TORSIONALLY RESILIENT COUPLING | Replace coupling or coupling element(s)                          | X      | WM00925 |
| QL4                                                     | 42000             | 18 YR | VALVE GEAR                               | Check rocker arm and valve bridge for wear, replace as necessary |        | WM00936 |
| QL4                                                     | 42000             | 18 YR | EXHAUST SYSTEM                           | Replace exhaust pipes                                            |        | WM01030 |
| DAY = Days<br>MON = Months<br>YR = Years<br>HRS = Hours |                   |       |                                          |                                                                  |        |         |

## 1.6 Maintenance Schedule Matrix 36000 HRS

### 500 to 15000 Operating hours

| Task    | Limit | Operating hours [HRS] |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 500                   | 1000 | 1500 | 2000 | 2500 | 3000 | 3500 | 4000 | 4500 | 5000 | 5500 | 6000 | 6500 | 7000 | 7500 | 8000 | 8500 | 9000 | 9500 | 10000 | 10500 | 11000 | 11500 | 12000 | 12500 | 13000 | 13500 | 14000 | 14500 | 15000 |
| WM00285 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00286 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00287 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00288 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00289 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00290 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00291 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00881 | 1 MON |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00067 | 2 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00116 | 2 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00126 | 5 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00128 | 5 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00063 | 2 YR  | X                     | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00066 | 9 YR  | X                     | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00149 | 1 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00084 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00086 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00170 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00050 | 1 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00136 | 1 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00076 | 3 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00664 | 3 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00692 | 5 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00052 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00094 | 18 YR |                       | X    |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00961 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00962 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00119 | 3 YR  |                       |      |      |      |      |      |      |      |      |      |      | X    |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00110 | 9 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | X    |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00090 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | X    |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00098 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      | X    |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00283 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       | X     |       |       |       |       |       |       |
| WM00976 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       | X     |       |       |       |       |       |       |
| WM00008 | 6 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00065 | 9 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00054 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00056 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00062 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00093 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00096 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00921 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |

DAY = Days

MON = Months

YR = Years

HRS = Hours

| Task    | Limit | Operating hours [HRS] |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 500                   | 1000 | 1500 | 2000 | 2500 | 3000 | 3500 | 4000 | 4500 | 5000 | 5500 | 6000 | 6500 | 7000 | 7500 | 8000 | 8500 | 9000 | 9500 | 10000 | 10500 | 11000 | 11500 | 12000 | 12500 | 13000 | 13500 | 14000 | 14500 | 15000 |
| WM00970 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00971 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00973 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00974 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00981 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00006 | 6 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00011 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00072 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00087 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00091 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00980 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00009 | 15 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00001 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00002 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00045 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00061 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00070 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00088 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00092 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00095 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00099 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00100 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00103 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00104 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00108 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00111 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00127 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00137 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00139 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00353 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00439 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00925 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00936 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM01030 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

### 15500 to 30000 Operating hours

| Task    | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 15500                 | 16000 | 16500 | 17000 | 17500 | 18000 | 18500 | 19000 | 19500 | 20000 | 20500 | 21000 | 21500 | 22000 | 22500 | 23000 | 23500 | 24000 | 24500 | 25000 | 25500 | 26000 | 26500 | 27000 | 27500 | 28000 | 28500 | 29000 | 29500 | 30000 |
| WM00285 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00286 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

| Task    | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 15500                 | 16000 | 16500 | 17000 | 17500 | 18000 | 18500 | 19000 | 19500 | 20000 | 20500 | 21000 | 21500 | 22000 | 22500 | 23000 | 23500 | 24000 | 24500 | 25000 | 25500 | 26000 | 26500 | 27000 | 27500 | 28000 | 28500 | 29000 | 29500 | 30000 |
| WM00287 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00288 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00289 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00290 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00291 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00881 | 1 MON |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00067 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00116 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00126 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00128 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00063 | 2 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00066 | 9 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00149 | 1 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00084 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00086 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00170 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00050 | 1 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |
| WM00136 | 1 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |
| WM00076 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |
| WM00664 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |
| WM00692 | 5 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |
| WM00052 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |
| WM00094 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |
| WM00961 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |
| WM00962 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |
| WM00119 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00110 | 9 YR  |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |
| WM00090 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |
| WM00098 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |
| WM00283 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00976 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00008 | 6 YR  |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00065 | 9 YR  |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00054 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00056 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00062 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00093 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00096 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00921 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00970 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00971 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00973 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00974 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00981 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00006 | 6 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |

DAY = Days  
 MON = Months  
 YR = Years  
 HRS = Hours

| Task         | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|--------------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|              |       | 15500                 | 16000 | 16500 | 17000 | 17500 | 18000 | 18500 | 19000 | 19500 | 20000 | 20500 | 21000 | 21500 | 22000 | 22500 | 23000 | 23500 | 24000 | 24500 | 25000 | 25500 | 26000 | 26500 | 27000 | 27500 | 28000 | 28500 | 29000 | 29500 | 30000 |
| WM00011      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00072      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00087      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00091      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00980      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00009      | 15 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00001      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00002      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00045      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00061      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00070      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00088      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00092      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00095      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00099      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00100      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00103      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00104      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00108      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00111      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00127      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00137      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00139      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00353      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00439      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00925      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00936      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM01030      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| DAY = Days   |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| MON = Months |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| YR = Years   |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| HRS = Hours  |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |

## 30500 to 36000 Operating hours

| Task         | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|--------------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
|              |       | 30500                 | 31000 | 31500 | 32000 | 32500 | 33000 | 33500 | 34000 | 34500 | 35000 | 35500 | 36000 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00285      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00286      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00287      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00288      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00289      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00290      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00291      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00881      | 1 MON |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| DAY = Days   |       |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| MON = Months |       |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| YR = Years   |       |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| HRS = Hours  |       |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

TIM-ID: 0000000000 - 000

TIM-ID: 000000000 - 000

| Task    | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
|---------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|--|--|--|--|
|         |       | 30500                 | 31000 | 31500 | 32000 | 32500 | 33000 | 33500 | 34000 | 34500 | 35000 | 35500 | 36000 |  |  |  |  |  |  |
| WM00067 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00116 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00126 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00128 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00063 | 2 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |  |  |  |  |  |  |
| WM00066 | 9 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |  |  |  |  |  |  |
| WM00149 | 1 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |  |  |  |  |  |  |
| WM00084 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |  |  |  |  |  |  |
| WM00086 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |  |  |  |  |  |  |
| WM00170 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |  |  |  |  |  |  |
| WM00050 | 1 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00136 | 1 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00076 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00664 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00692 | 5 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00052 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00094 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00961 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00962 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00119 | 3 YR  |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00110 | 9 YR  |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00090 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00098 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00283 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00976 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00008 | 6 YR  |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00065 | 9 YR  |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00054 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00056 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00062 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00093 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00096 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00921 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00970 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00971 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00973 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00974 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00981 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00006 | 6 YR  |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00011 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00072 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00087 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00091 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00980 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00009 | 15 YR |                       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |

DAY = Days  
 MON = Months  
 YR = Years  
 HRS = Hours

| Task    | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
|---------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|--|--|--|--|--|--|--|--|--|--|--|--|--|
|         |       | 30500                 | 31000 | 31500 | 32000 | 32500 | 33000 | 33500 | 34000 | 34500 | 35000 | 35500 | 36000 |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00001 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00002 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00045 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00061 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00070 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00088 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00092 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00095 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00099 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00100 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00103 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00104 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00108 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00111 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00127 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00137 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00139 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00353 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00439 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00925 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00936 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |
| WM01030 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |  |  |  |  |  |  |  |  |  |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

## 1.7 Cyclical maintenance tasks 36000 HRS

| Qualification | Interval [HRS] | Limit | Item                                                               | Maintenance tasks                                                                                                                                                                          | Option | Task    |
|---------------|----------------|-------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| <b>Engine</b> |                |       |                                                                    |                                                                                                                                                                                            |        |         |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check engine oil level                                                                                                                                                                     |        | WM00285 |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Visually inspect for leaks and general condition of engine                                                                                                                                 |        | WM00286 |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check maintenance indicator of air filter                                                                                                                                                  | X      | WM00287 |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check relief bore of coolant pump                                                                                                                                                          |        | WM00288 |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | When engine is running, check for abnormal running noises, exhaust gas color and vibrations                                                                                                |        | WM00289 |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Drain water and contaminants from fuel prefilter                                                                                                                                           | X      | WM00290 |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check pointer of vacuum gauge of fuel prefilter                                                                                                                                            | X      | WM00291 |
| QL1           | -              | 1 MON | BENDING / TORSIONALLY RESILIENT COUPLING AND STEEL-SPRING COUPLING | Check coupling (visual inspection), replace as necessary                                                                                                                                   | X      | WM00881 |
| QL1           | -              | 2 YR  | LUBE OIL SYSTEM                                                    | Replace oil filter at every engine oil change, or when the time limit (years) is reached, at the latest                                                                                    | X      | WM00067 |
| QL1           | -              | 2 YR  | COOLANT SYSTEM                                                     | Replace filter                                                                                                                                                                             |        | WM00116 |
| QL1           | -              | 5 YR  | ENGINE MOUNTING                                                    | Measure height of rubber element                                                                                                                                                           |        | WM00126 |
| QL1           | -              | 5 YR  | ENGINE MOUNTING                                                    | Measure buffer play, adjust if necessary                                                                                                                                                   |        | WM00128 |
| QL1           | 500            | 2 YR  | LUBE OIL SYSTEM                                                    | Automatic centrifugal oil filter, check layer thickness of oil residue, clean, and replace paper element at every engine oil change at the latest                                          |        | WM00063 |
| QL1           | 500            | 9 YR  | AUTOMATIC OIL FILTER                                               | Automatic oil filter, check and clean filtering screen                                                                                                                                     | X      | WM00066 |
| QL1           | 1000           | 1 YR  | AIR SUPPLY SYSTEM                                                  | Clean air filter, empty dust collector box                                                                                                                                                 | X      | WM00149 |
| QL1           | 1000           | 2 YR  | FUEL SYSTEM (LOW PRESSURE)                                         | Fuel prefilter, replace filter                                                                                                                                                             | X      | WM00084 |
| QL1           | 1000           | 2 YR  | FUEL SYSTEM (LOW PRESSURE)                                         | Replace easy-change fuel filter                                                                                                                                                            |        | WM00086 |
| QL1           | 1000           | 2 YR  | AUXILIARY POWER TAKE-OFF                                           | Check belt condition, replace if necessary. Adjust tension.                                                                                                                                | X      | WM00170 |
| QL1           | 3000           | 1 YR  | ENGINE MOUNTING                                                    | Check general condition of rubber mounts (visual inspection)                                                                                                                               |        | WM00050 |
| QL1           | 3000           | 1 YR  | ENGINE MOUNTING                                                    | Check securing screws for secure seating                                                                                                                                                   |        | WM00136 |
| QL1           | 3000           | 3 YR  | AIR SUPPLY SYSTEM                                                  | Replace air filter                                                                                                                                                                         | X      | WM00076 |
| QL1           | 3000           | 3 YR  | COOLANT SYSTEM                                                     | Replace valve cover                                                                                                                                                                        | X      | WM00664 |
| QL1           | 3000           | 5 YR  | CRANKCASE BREATHER                                                 | Replace filter or filter element (HE200)                                                                                                                                                   | X      | WM00692 |
| QL1           | 3000           | 18 YR | BATTERY-CHARGING GENERATOR                                         | Check condition of coupling                                                                                                                                                                | X      | WM00052 |
| QL1           | 3000           | 18 YR | VALVE GEAR                                                         | Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul. |        | WM00094 |
| QL1           | 3000           | 18 YR | GROUNDING SYSTEM                                                   | Replace carbon brushes at flywheel                                                                                                                                                         | X      | WM00961 |
| QL1           | 3000           | 18 YR | GROUNDING SYSTEM                                                   | Replace carbon brushes and cylindrical roller bearings of grounding equipment at vibration damper                                                                                          | X      | WM00962 |
| QL1           | 6000           | 3 YR  | CRANKCASE BREATHER                                                 | Replace filter or filter element                                                                                                                                                           | X      | WM00119 |
| QL1           | 9000           | 9 YR  | RUNNING GEAR                                                       | Inspect cylinder liner with endoscope                                                                                                                                                      |        | WM00110 |
| QL1           | 9000           | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace fuel injection valve/injector                                                                                                                                                      |        | WM00090 |

DAY = Days  
 MON = Months  
 YR = Years  
 HRS = Hours

| Qualification | Interval [HRS] | Limit | Item                                                               | Maintenance tasks                                                                                                          | Option | Task    |
|---------------|----------------|-------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------|---------|
| QL1           | 9000           | 18 YR | CYLINDER HEAD                                                      | Measure valve protrusion                                                                                                   |        | WM00098 |
| QL3           | 12000          | 18 YR | AUXILIARY POWER TAKE-OFF                                           | Replace auxiliary power take-off                                                                                           | X      | WM00283 |
| QL3           | 12000          | 18 YR | BATTERY-CHARGING GENERATOR                                         | Overhaul or replace battery-charging generator                                                                             | X      | WM00976 |
| QL3           | 18000          | 6 YR  | RUBBER SLEEVE                                                      | Replace rubber sleeve                                                                                                      |        | WM00008 |
| QL3           | 18000          | 9 YR  | AUTOMATIC OIL FILTER                                               | Automatic oil filter, remove and clean filter housing, replace filter element and bearing                                  | X      | WM00065 |
| QL3           | 18000          | 18 YR | COOLANT SYSTEM                                                     | Check LT thermostat housing and replace thermal actuator                                                                   |        | WM00054 |
| QL3           | 18000          | 18 YR | COOLANT SYSTEM                                                     | Check HT thermostat housing and replace thermal actuator                                                                   |        | WM00056 |
| QL3           | 18000          | 18 YR | LUBE OIL SYSTEM                                                    | Check centrifugal oil filter for wear                                                                                      |        | WM00062 |
| QL3           | 18000          | 18 YR | VALVE GEAR                                                         | Check rocker arm and valve bridge for wear. Visually inspect swing follower with endoscope through pushrod bore            |        | WM00093 |
| QL3           | 18000          | 18 YR | VALVE GEAR                                                         | Visually inspect camshaft running surfaces with endoscope through pushrod bore                                             |        | WM00096 |
| QL3           | 18000          | 18 YR | COOLANT SYSTEM                                                     | Replace sealing material for installation of coolant pump(s) and thermostat(s) (only LT and HT)                            |        | WM00921 |
| QL3           | 18000          | 18 YR | EXHAUST TURBOCHARGER                                               | Overhaul or replace exhaust turbocharger                                                                                   |        | WM00970 |
| QL3           | 18000          | 18 YR | STARTING SYSTEM                                                    | Overhaul or replace starter                                                                                                |        | WM00971 |
| QL3           | 18000          | 18 YR | COOLANT SYSTEM                                                     | Overhaul or replace coolant pump in LT circuit                                                                             |        | WM00973 |
| QL3           | 18000          | 18 YR | COOLANT SYSTEM                                                     | Overhaul or replace coolant pump in HT circuit                                                                             |        | WM00974 |
| QL3           | 18000          | 18 YR | LUBE OIL SYSTEM                                                    | Clean oil cooler and check for leaks or replace                                                                            |        | WM00981 |
| QL3           | 36000          | 6 YR  | HOSE                                                               | Replace hose line                                                                                                          |        | WM00006 |
| QL3           | 36000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace HP fuel sensor                                                                                                     |        | WM00011 |
| QL3           | 36000          | 18 YR | EXHAUST SYSTEM                                                     | Replace exhaust pipe bellows                                                                                               |        | WM00072 |
| QL3           | 36000          | 18 YR | FUEL SYSTEM (LOW PRESSURE)                                         | Replace pump                                                                                                               |        | WM00087 |
| QL3           | 36000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace HP pump                                                                                                            |        | WM00091 |
| QL3           | 36000          | 18 YR | CYLINDER HEAD                                                      | Overhaul or replace cylinder head                                                                                          |        | WM00980 |
| QL4           | -              | 15 YR | STARTING SYSTEM                                                    | Replace CAPOS                                                                                                              | X      | WM00009 |
| QL4           | 36000          | 18 YR | ENGINE, GENERAL                                                    | Replace elastomer parts and seals                                                                                          |        | WM00001 |
| QL4           | 36000          | 18 YR | ENGINE, GENERAL                                                    | Disassemble engine completely. Check engine components according to assembly instructions, replace or repair as necessary. |        | WM00002 |
| QL4           | 36000          | 18 YR | BENDING / TORSIONALLY RESILIENT COUPLING AND STEEL-SPRING COUPLING | Check coupling, replace if necessary                                                                                       | X      | WM00045 |
| QL4           | 36000          | 18 YR | OIL REPLENISHING UNIT                                              | Replace gear pump                                                                                                          | X      | WM00061 |
| QL4           | 36000          | 18 YR | LUBE OIL SYSTEM                                                    | Check pump, replace if necessary                                                                                           |        | WM00070 |
| QL4           | 36000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace pressure relief valve                                                                                              |        | WM00088 |
| QL4           | 36000          | 18 YR | VALVE GEAR                                                         | Replace swing follower and swing follower shaft                                                                            |        | WM00092 |
| QL4           | 36000          | 18 YR | VALVE GEAR                                                         | Replace thrust bearing flange                                                                                              |        | WM00095 |
| QL4           | 36000          | 18 YR | RUNNING GEAR                                                       | Replace conrod bearings                                                                                                    |        | WM00099 |
| QL4           | 36000          | 18 YR | RUNNING GEAR                                                       | Replace rings                                                                                                              |        | WM00100 |
| QL4           | 36000          | 18 YR | RUNNING GEAR                                                       | Replace crankshaft bearings                                                                                                |        | WM00103 |
| QL4           | 36000          | 18 YR | GEAR TRAIN                                                         | Check tooth flanks in gear train for wear (visual inspection), replace bushing                                             |        | WM00104 |
| QL4           | 36000          | 18 YR | RUNNING GEAR                                                       | Replace cylinder liners                                                                                                    |        | WM00108 |
| QL4           | 36000          | 18 YR | CAMSHAFT                                                           | Replace camshaft bearings                                                                                                  |        | WM00111 |
| QL4           | 36000          | 18 YR | ENGINE MOUNTING                                                    | Replace rubber element of rubber mounts                                                                                    |        | WM00127 |

DAY = Days  
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TIM-ID: 0000000000 - 000

| Qualification                                           | Interval [HRS] | Limit | Item                                     | Maintenance tasks                                                | Option | Task    |
|---------------------------------------------------------|----------------|-------|------------------------------------------|------------------------------------------------------------------|--------|---------|
| QL4                                                     | 36000          | 18 YR | SENSOR TECHNOLOGY AND WIRING HARNESS     | Replace wiring harness                                           |        | WM00137 |
| QL4                                                     | 36000          | 18 YR | GROUNDING SYSTEM                         | Check flywheel (insulated), overhaul if necessary                | X      | WM00139 |
| QL4                                                     | 36000          | 18 YR | RUNNING GEAR                             | Replace conrod                                                   |        | WM00353 |
| QL4                                                     | 36000          | 18 YR | RUNNING GEAR                             | Replace vibration damper                                         |        | WM00439 |
| QL4                                                     | 36000          | 18 YR | BENDING / TORSIONALLY RESILIENT COUPLING | Replace coupling or coupling element(s)                          | X      | WM00925 |
| QL4                                                     | 36000          | 18 YR | VALVE GEAR                               | Check rocker arm and valve bridge for wear, replace as necessary |        | WM00936 |
| QL4                                                     | 36000          | 18 YR | EXHAUST SYSTEM                           | Replace exhaust pipes                                            |        | WM01030 |
| DAY = Days<br>MON = Months<br>YR = Years<br>HRS = Hours |                |       |                                          |                                                                  |        |         |

## 1.8 Maintenance Schedule Matrix 30000 HRS

### 500 to 15000 Operating hours

| Task    | Limit | Operating hours [HRS] |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 500                   | 1000 | 1500 | 2000 | 2500 | 3000 | 3500 | 4000 | 4500 | 5000 | 5500 | 6000 | 6500 | 7000 | 7500 | 8000 | 8500 | 9000 | 9500 | 10000 | 10500 | 11000 | 11500 | 12000 | 12500 | 13000 | 13500 | 14000 | 14500 | 15000 |
| WM00285 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00286 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00287 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00288 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00289 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00290 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00291 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00881 | 1 MON |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00067 | 2 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00116 | 2 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00126 | 5 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00128 | 5 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00063 | 2 YR  | X                     | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00066 | 9 YR  | X                     | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00149 | 1 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00084 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00086 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00170 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00050 | 1 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00136 | 1 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00076 | 3 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00664 | 3 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00692 | 5 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00052 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00094 | 18 YR |                       | X    |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00961 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00962 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00119 | 3 YR  |                       |      |      |      |      |      |      |      |      |      |      | X    |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00110 | 9 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      | X    |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00090 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      | X    |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00098 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      | X    |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00008 | 6 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00065 | 9 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00054 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00056 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00062 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00093 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00096 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00283 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00921 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00970 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |

DAY = Days

MON = Months

YR = Years

HRS = Hours

TIM-ID: 0000000000 - 000

| Task         | Limit | Operating hours [HRS] |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
|--------------|-------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|              |       | 500                   | 1000 | 1500 | 2000 | 2500 | 3000 | 3500 | 4000 | 4500 | 5000 | 5500 | 6000 | 6500 | 7000 | 7500 | 8000 | 8500 | 9000 | 9500 | 10000 | 10500 | 11000 | 11500 | 12000 | 12500 | 13000 | 13500 | 14000 | 14500 | 15000 |
| WM00971      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00973      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00974      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00976      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00981      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00006      | 6 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00011      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00072      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00087      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00091      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00980      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00009      | 15 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00001      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00002      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00045      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       | X     |
| WM00061      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00070      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00088      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00092      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00095      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00099      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00100      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00103      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00104      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00108      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00111      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00127      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00137      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00139      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00353      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00439      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00925      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00936      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM01030      | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| DAY = Days   |       |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| MON = Months |       |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| YR = Years   |       |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| HRS = Hours  |       |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |

## 15500 to 30000 Operating hours

| Task         | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|--------------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|              |       | 15500                 | 16000 | 16500 | 17000 | 17500 | 18000 | 18500 | 19000 | 19500 | 20000 | 20500 | 21000 | 21500 | 22000 | 22500 | 23000 | 23500 | 24000 | 24500 | 25000 | 25500 | 26000 | 26500 | 27000 | 27500 | 28000 | 28500 | 29000 | 29500 | 30000 |
| WM00285      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00286      | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| DAY = Days   |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| MON = Months |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| YR = Years   |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| HRS = Hours  |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |

| Task    | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 15500                 | 16000 | 16500 | 17000 | 17500 | 18000 | 18500 | 19000 | 19500 | 20000 | 20500 | 21000 | 21500 | 22000 | 22500 | 23000 | 23500 | 24000 | 24500 | 25000 | 25500 | 26000 | 26500 | 27000 | 27500 | 28000 | 28500 | 29000 | 29500 | 30000 |
| WM00287 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00288 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00289 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00290 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00291 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00881 | 1 MON |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00067 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00116 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00126 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00128 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00063 | 2 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00066 | 9 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00149 | 1 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00084 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00086 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00170 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00050 | 1 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |
| WM00136 | 1 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |
| WM00076 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |
| WM00664 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |
| WM00692 | 5 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |
| WM00052 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |
| WM00094 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |
| WM00961 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |
| WM00962 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |
| WM00119 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00110 | 9 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00090 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00098 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00008 | 6 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00065 | 9 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00054 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00056 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00062 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00093 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00096 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00283 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00921 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00970 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00971 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00973 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00974 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00976 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00981 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |
| WM00006 | 6 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

| Task         | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |
|--------------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|---|
|              |       | 15500                 | 16000 | 16500 | 17000 | 17500 | 18000 | 18500 | 19000 | 19500 | 20000 | 20500 | 21000 | 21500 | 22000 | 22500 | 23000 | 23500 | 24000 | 24500 | 25000 | 25500 | 26000 | 26500 | 27000 | 27500 | 28000 | 28500 | 29000 | 29500 | 30000 |   |
| WM00011      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00072      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00087      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00091      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00980      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00009      | 15 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |
| WM00001      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00002      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00045      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00061      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00070      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00088      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00092      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00095      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00099      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00100      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00103      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00104      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00108      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00111      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00127      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00137      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00139      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00353      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00439      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00925      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM00936      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| WM01030      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X |
| DAY = Days   |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |
| MON = Months |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |
| YR = Years   |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |
| HRS = Hours  |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |

## 1.9 Cyclical maintenance tasks 30000 HRS

| Qualification | Interval [HRS] | Limit | Item                                                               | Maintenance tasks                                                                                                                                                                          | Option | Task    |
|---------------|----------------|-------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| <b>Engine</b> |                |       |                                                                    |                                                                                                                                                                                            |        |         |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check engine oil level                                                                                                                                                                     |        | WM00285 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Visually inspect for leaks and general condition of engine                                                                                                                                 |        | WM00286 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check maintenance indicator of air filter                                                                                                                                                  | X      | WM00287 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check relief bore of coolant pump                                                                                                                                                          |        | WM00288 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | When engine is running, check for abnormal running noises, exhaust gas color and vibrations                                                                                                |        | WM00289 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Drain water and contaminants from fuel prefilter                                                                                                                                           | X      | WM00290 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check pointer of vacuum gauge of fuel prefilter                                                                                                                                            | X      | WM00291 |
| QL 1          | -              | 1 MON | BENDING / TORSIONALLY RESILIENT COUPLING AND STEEL-SPRING COUPLING | Check coupling (visual inspection), replace as necessary                                                                                                                                   | X      | WM00881 |
| QL 1          | -              | 2 YR  | LUBE OIL SYSTEM                                                    | Replace oil filter at every engine oil change, or when the time limit (years) is reached, at the latest                                                                                    | X      | WM00067 |
| QL 1          | -              | 2 YR  | COOLANT SYSTEM                                                     | Replace filter                                                                                                                                                                             |        | WM00116 |
| QL 1          | -              | 5 YR  | ENGINE MOUNTING                                                    | Measure height of rubber element                                                                                                                                                           |        | WM00126 |
| QL 1          | -              | 5 YR  | ENGINE MOUNTING                                                    | Measure buffer play, adjust if necessary                                                                                                                                                   |        | WM00128 |
| QL 1          | 500            | 2 YR  | LUBE OIL SYSTEM                                                    | Automatic centrifugal oil filter, check layer thickness of oil residue, clean, and replace paper element at every engine oil change at the latest                                          |        | WM00063 |
| QL 1          | 500            | 9 YR  | AUTOMATIC OIL FILTER                                               | Automatic oil filter, check and clean filtering screen                                                                                                                                     | X      | WM00066 |
| QL 1          | 1000           | 1 YR  | AIR SUPPLY SYSTEM                                                  | Clean air filter, empty dust collector box                                                                                                                                                 | X      | WM00149 |
| QL 1          | 1000           | 2 YR  | FUEL SYSTEM (LOW PRESSURE)                                         | Fuel prefilter, replace filter                                                                                                                                                             | X      | WM00084 |
| QL 1          | 1000           | 2 YR  | FUEL SYSTEM (LOW PRESSURE)                                         | Replace easy-change fuel filter                                                                                                                                                            |        | WM00086 |
| QL 1          | 1000           | 2 YR  | AUXILIARY POWER TAKE-OFF                                           | Check belt condition, replace if necessary. Adjust tension.                                                                                                                                | X      | WM00170 |
| QL 1          | 3000           | 1 YR  | ENGINE MOUNTING                                                    | Check general condition of rubber mounts (visual inspection)                                                                                                                               |        | WM00050 |
| QL 1          | 3000           | 1 YR  | ENGINE MOUNTING                                                    | Check securing screws for secure seating                                                                                                                                                   |        | WM00136 |
| QL 1          | 3000           | 3 YR  | AIR SUPPLY SYSTEM                                                  | Replace air filter                                                                                                                                                                         | X      | WM00076 |
| QL 1          | 3000           | 3 YR  | COOLANT SYSTEM                                                     | Replace valve cover                                                                                                                                                                        | X      | WM00664 |
| QL 1          | 3000           | 5 YR  | CRANKCASE BREATHER                                                 | Replace filter or filter element (HE200)                                                                                                                                                   | X      | WM00692 |
| QL 1          | 3000           | 18 YR | BATTERY-CHARGING GENERATOR                                         | Check condition of coupling                                                                                                                                                                | X      | WM00052 |
| QL 1          | 3000           | 18 YR | VALVE GEAR                                                         | Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul. |        | WM00094 |
| QL 1          | 3000           | 18 YR | GROUNDING SYSTEM                                                   | Replace carbon brushes at flywheel                                                                                                                                                         | X      | WM00961 |
| QL 1          | 3000           | 18 YR | GROUNDING SYSTEM                                                   | Replace carbon brushes and cylindrical roller bearings of grounding equipment at vibration damper                                                                                          | X      | WM00962 |
| QL 1          | 6000           | 3 YR  | CRANKCASE BREATHER                                                 | Replace filter or filter element                                                                                                                                                           | X      | WM00119 |
| QL 1          | 7500           | 9 YR  | RUNNING GEAR                                                       | Inspect cylinder liner with endoscope                                                                                                                                                      |        | WM00110 |
| QL 1          | 7500           | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace fuel injection valve/injector                                                                                                                                                      |        | WM00090 |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

TIM-ID: 0000000000 - 000

| Qualification | Interval [HRS] | Limit | Item                                                               | Maintenance tasks                                                                                                          | Option | Task    |
|---------------|----------------|-------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------|---------|
| QL1           | 7500           | 18 YR | CYLINDER HEAD                                                      | Measure valve protrusion                                                                                                   |        | WM00098 |
| QL3           | 15000          | 6 YR  | RUBBER SLEEVE                                                      | Replace rubber sleeve                                                                                                      |        | WM00008 |
| QL3           | 15000          | 9 YR  | AUTOMATIC OIL FILTER                                               | Automatic oil filter, remove and clean filter housing, replace filter element and bearing                                  | X      | WM00065 |
| QL3           | 15000          | 18 YR | COOLANT SYSTEM                                                     | Check LT thermostat housing and replace thermal actuator                                                                   |        | WM00054 |
| QL3           | 15000          | 18 YR | COOLANT SYSTEM                                                     | Check HT thermostat housing and replace thermal actuator                                                                   |        | WM00056 |
| QL3           | 15000          | 18 YR | LUBE OIL SYSTEM                                                    | Check centrifugal oil filter for wear                                                                                      |        | WM00062 |
| QL3           | 15000          | 18 YR | VALVE GEAR                                                         | Check rocker arm and valve bridge for wear. Visually inspect swing follower with endoscope through pushrod bore            |        | WM00093 |
| QL3           | 15000          | 18 YR | VALVE GEAR                                                         | Visually inspect camshaft running surfaces with endoscope through pushrod bore                                             |        | WM00096 |
| QL3           | 15000          | 18 YR | AUXILIARY POWER TAKE-OFF                                           | Replace auxiliary power take-off                                                                                           | X      | WM00283 |
| QL3           | 15000          | 18 YR | COOLANT SYSTEM                                                     | Replace sealing material for installation of coolant pump(s) and thermostat(s) (only LT and HT)                            |        | WM00921 |
| QL3           | 15000          | 18 YR | EXHAUST TURBOCHARGER                                               | Overhaul or replace exhaust turbocharger                                                                                   |        | WM00970 |
| QL3           | 15000          | 18 YR | STARTING SYSTEM                                                    | Overhaul or replace starter                                                                                                |        | WM00971 |
| QL3           | 15000          | 18 YR | COOLANT SYSTEM                                                     | Overhaul or replace coolant pump in LT circuit                                                                             |        | WM00973 |
| QL3           | 15000          | 18 YR | COOLANT SYSTEM                                                     | Overhaul or replace coolant pump in HT circuit                                                                             |        | WM00974 |
| QL3           | 15000          | 18 YR | BATTERY-CHARGING GENERATOR                                         | Overhaul or replace battery-charging generator                                                                             | X      | WM00976 |
| QL3           | 15000          | 18 YR | LUBE OIL SYSTEM                                                    | Clean oil cooler and check for leaks or replace                                                                            |        | WM00981 |
| QL3           | 30000          | 6 YR  | HOSE                                                               | Replace hose line                                                                                                          |        | WM00006 |
| QL3           | 30000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace HP fuel sensor                                                                                                     |        | WM00011 |
| QL3           | 30000          | 18 YR | EXHAUST SYSTEM                                                     | Replace exhaust pipe bellows                                                                                               |        | WM00072 |
| QL3           | 30000          | 18 YR | FUEL SYSTEM (LOW PRESSURE)                                         | Replace pump                                                                                                               |        | WM00087 |
| QL3           | 30000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace HP pump                                                                                                            |        | WM00091 |
| QL3           | 30000          | 18 YR | CYLINDER HEAD                                                      | Overhaul or replace cylinder head                                                                                          |        | WM00980 |
| QL4           | -              | 15 YR | STARTING SYSTEM                                                    | Replace CAPOS                                                                                                              | X      | WM00009 |
| QL4           | 30000          | 18 YR | ENGINE, GENERAL                                                    | Replace elastomer parts and seals                                                                                          |        | WM00001 |
| QL4           | 30000          | 18 YR | ENGINE, GENERAL                                                    | Disassemble engine completely. Check engine components according to assembly instructions, replace or repair as necessary. |        | WM00002 |
| QL4           | 30000          | 18 YR | BENDING / TORSIONALLY RESILIENT COUPLING AND STEEL-SPRING COUPLING | Check coupling, replace if necessary                                                                                       | X      | WM00045 |
| QL4           | 30000          | 18 YR | OIL REPLENISHING UNIT                                              | Replace gear pump                                                                                                          | X      | WM00061 |
| QL4           | 30000          | 18 YR | LUBE OIL SYSTEM                                                    | Check pump, replace if necessary                                                                                           |        | WM00070 |
| QL4           | 30000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace pressure relief valve                                                                                              |        | WM00088 |
| QL4           | 30000          | 18 YR | VALVE GEAR                                                         | Replace swing follower and swing follower shaft                                                                            |        | WM00092 |
| QL4           | 30000          | 18 YR | VALVE GEAR                                                         | Replace thrust bearing flange                                                                                              |        | WM00095 |
| QL4           | 30000          | 18 YR | RUNNING GEAR                                                       | Replace conrod bearings                                                                                                    |        | WM00099 |
| QL4           | 30000          | 18 YR | RUNNING GEAR                                                       | Replace rings                                                                                                              |        | WM00100 |
| QL4           | 30000          | 18 YR | RUNNING GEAR                                                       | Replace crankshaft bearings                                                                                                |        | WM00103 |
| QL4           | 30000          | 18 YR | GEAR TRAIN                                                         | Check tooth flanks in gear train for wear (visual inspection), replace bushing                                             |        | WM00104 |
| QL4           | 30000          | 18 YR | RUNNING GEAR                                                       | Replace cylinder liners                                                                                                    |        | WM00108 |
| QL4           | 30000          | 18 YR | CAMSHAFT                                                           | Replace camshaft bearings                                                                                                  |        | WM00111 |
| QL4           | 30000          | 18 YR | ENGINE MOUNTING                                                    | Replace rubber element of rubber mounts                                                                                    |        | WM00127 |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

| Qualification                                           | Interval<br>[HRS] | Limit | Item                                     | Maintenance tasks                                                | Option | Task    |
|---------------------------------------------------------|-------------------|-------|------------------------------------------|------------------------------------------------------------------|--------|---------|
| QL4                                                     | 30000             | 18 YR | SENSOR TECHNOLOGY AND WIRING HARNESS     | Replace wiring harness                                           |        | WM00137 |
| QL4                                                     | 30000             | 18 YR | GROUNDING SYSTEM                         | Check flywheel (insulated), overhaul if necessary                | X      | WM00139 |
| QL4                                                     | 30000             | 18 YR | RUNNING GEAR                             | Replace conrod                                                   |        | WM00353 |
| QL4                                                     | 30000             | 18 YR | RUNNING GEAR                             | Replace vibration damper                                         |        | WM00439 |
| QL4                                                     | 30000             | 18 YR | BENDING / TORSIONALLY RESILIENT COUPLING | Replace coupling or coupling element(s)                          | X      | WM00925 |
| QL4                                                     | 30000             | 18 YR | VALVE GEAR                               | Check rocker arm and valve bridge for wear, replace as necessary |        | WM00936 |
| QL4                                                     | 30000             | 18 YR | EXHAUST SYSTEM                           | Replace exhaust pipes                                            |        | WM01030 |
| DAY = Days<br>MON = Months<br>YR = Years<br>HRS = Hours |                   |       |                                          |                                                                  |        |         |

## 1.10 Maintenance Schedule Matrix 27000 HRS

### 500 to 14500 Operating hours

| Task    | Limit | Operating hours [HRS] |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 500                   | 1000 | 1500 | 2000 | 2500 | 3000 | 3500 | 4000 | 4500 | 5000 | 5500 | 6000 | 6500 | 6750 | 7000 | 7500 | 8000 | 8500 | 9000 | 9500 | 10000 | 10500 | 11000 | 11500 | 12000 | 12500 | 13000 | 13500 | 14000 | 14500 |
| WM00285 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00286 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00287 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00288 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00289 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00290 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00291 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00881 | 1 MON |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00067 | 2 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00116 | 2 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00126 | 5 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00128 | 5 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00063 | 2 YR  | X                     | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    |      | X    | X    | X    | X    | X    | X    | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00066 | 9 YR  | X                     | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    |      | X    | X    | X    | X    | X    | X    | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00149 | 1 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       |
| WM00084 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       |
| WM00086 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       |
| WM00170 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       |
| WM00050 | 1 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      |      | X    |      |       |       |       |       | X     |       |       |       |       |       |
| WM00136 | 1 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      |      | X    |      |       |       |       |       | X     |       |       |       |       |       |
| WM00076 | 3 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      |      | X    |      |       |       |       |       | X     |       |       |       |       |       |
| WM00664 | 3 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      |      | X    |      |       |       |       |       | X     |       |       |       |       |       |
| WM00692 | 5 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      |      | X    |      |       |       |       |       | X     |       |       |       |       |       |
| WM00052 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      |      | X    |      |       |       |       |       | X     |       |       |       |       |       |
| WM00094 | 18 YR |                       | X    |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      |      | X    |      |       |       |       |       | X     |       |       |       |       |       |
| WM00961 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      |      | X    |      |       |       |       |       | X     |       |       |       |       |       |
| WM00962 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      |      | X    |      |       |       |       |       | X     |       |       |       |       |       |
| WM00119 | 3 YR  |                       |      |      |      |      |      |      |      | X    |      |      |      |      |      |      |      |      |      | X    |      |       |       |       |       |       |       |       |       | X     |       |
| WM00110 | 9 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      | X    |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |
| WM00090 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      | X    |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |
| WM00098 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      | X    |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |
| WM00008 | 6 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |
| WM00065 | 9 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |
| WM00054 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |
| WM00056 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |
| WM00062 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |
| WM00093 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |
| WM00096 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |
| WM00283 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |
| WM00921 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |
| WM00970 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       | X     |       |

DAY = Days

MON = Months

YR = Years

HRS = Hours

| Task    | Limit | Operating hours [HRS] |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 500                   | 1000 | 1500 | 2000 | 2500 | 3000 | 3500 | 4000 | 4500 | 5000 | 5500 | 6000 | 6500 | 6750 | 7000 | 7500 | 8000 | 8500 | 9000 | 9500 | 10000 | 10500 | 11000 | 11500 | 12000 | 12500 | 13000 | 13500 | 14000 | 14500 |
| WM00971 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       | X     |       |       |
| WM00973 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       | X     |       |       |
| WM00974 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       | X     |       |       |
| WM00976 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       | X     |       |       |
| WM00981 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       | X     |       |       |
| WM00006 | 6 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00011 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00072 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00087 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00091 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00980 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00009 | 15 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00001 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00002 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00045 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00061 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00070 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00088 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00092 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00095 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00099 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00100 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00103 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00104 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00108 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00111 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00127 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00137 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00139 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00353 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00439 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00925 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM00936 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |
| WM01030 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

### 15000 to 27000 Operating hours

| Task    | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 15000                 | 15500 | 16000 | 16500 | 17000 | 17500 | 18000 | 18500 | 19000 | 19500 | 20000 | 20250 | 20500 | 21000 | 21500 | 22000 |
| WM00285 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |
| WM00286 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

TIM-ID: 000000000 - 000

| Task    | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |  |  |  |
|---------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|---|--|--|--|
|         |       | 15000                 | 15500 | 16000 | 16500 | 17000 | 17500 | 18000 | 18500 | 19000 | 19500 | 20000 | 20250 | 20500 | 21000 | 21500 | 22000 | 22500 | 23000 | 23500 | 24000 | 24500 | 25000 | 25500 | 26000 | 26500 | 27000 |   |  |  |  |
| WM00287 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |  |  |  |
| WM00288 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |  |  |  |
| WM00289 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |  |  |  |
| WM00290 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |  |  |  |
| WM00291 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |  |  |  |
| WM00881 | 1 MON |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |  |  |  |
| WM00067 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |  |  |  |
| WM00116 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |  |  |  |
| WM00126 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |  |  |  |
| WM00128 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |   |  |  |  |
| WM00063 | 2 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |       | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X |  |  |  |
| WM00066 | 9 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |       | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X |  |  |  |
| WM00149 | 1 YR  | X                     |       | X     |       | X     |       | X     |       | X     |       | X     |       |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |   |  |  |  |
| WM00084 | 2 YR  | X                     |       | X     |       | X     |       | X     |       | X     |       | X     |       |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |   |  |  |  |
| WM00086 | 2 YR  | X                     |       | X     |       | X     |       | X     |       | X     |       | X     |       |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |   |  |  |  |
| WM00170 | 2 YR  | X                     |       | X     |       | X     |       | X     |       | X     |       | X     |       |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |   |  |  |  |
| WM00050 | 1 YR  | X                     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       |       |       | X     |   |  |  |  |
| WM00136 | 1 YR  | X                     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       |       |       | X     |   |  |  |  |
| WM00076 | 3 YR  | X                     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       |       |       | X     |   |  |  |  |
| WM00664 | 3 YR  | X                     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       |       |       | X     |   |  |  |  |
| WM00692 | 5 YR  | X                     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       |       |       | X     |   |  |  |  |
| WM00052 | 18 YR | X                     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       |       |       | X     |   |  |  |  |
| WM00094 | 18 YR | X                     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       |       |       | X     |   |  |  |  |
| WM00961 | 18 YR | X                     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       |       |       | X     |   |  |  |  |
| WM00962 | 18 YR | X                     |       |       |       |       |       | X     |       |       |       |       |       |       | X     |       |       |       |       | X     |       |       |       |       |       |       | X     |   |  |  |  |
| WM00119 | 3 YR  |                       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00110 | 9 YR  |                       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00090 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00098 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00008 | 6 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00065 | 9 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00054 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00056 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00062 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00093 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00096 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00283 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00921 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00970 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00971 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00973 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00974 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00976 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00981 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |
| WM00006 | 6 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |   |  |  |  |

DAY = Days  
 MON = Months  
 YR = Years  
 HRS = Hours



## 1.11 Cyclical maintenance tasks 27000 HRS

| Qualification | Interval [HRS] | Limit | Item                                                               | Maintenance tasks                                                                                                                                                                          | Option | Task    |
|---------------|----------------|-------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| <b>Engine</b> |                |       |                                                                    |                                                                                                                                                                                            |        |         |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check engine oil level                                                                                                                                                                     |        | WM00285 |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Visually inspect for leaks and general condition of engine                                                                                                                                 |        | WM00286 |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check maintenance indicator of air filter                                                                                                                                                  | X      | WM00287 |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check relief bore of coolant pump                                                                                                                                                          |        | WM00288 |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | When engine is running, check for abnormal running noises, exhaust gas color and vibrations                                                                                                |        | WM00289 |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Drain water and contaminants from fuel prefilter                                                                                                                                           | X      | WM00290 |
| QL1           | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check pointer of vacuum gauge of fuel prefilter                                                                                                                                            | X      | WM00291 |
| QL1           | -              | 1 MON | BENDING / TORSIONALLY RESILIENT COUPLING AND STEEL-SPRING COUPLING | Check coupling (visual inspection), replace as necessary                                                                                                                                   | X      | WM00881 |
| QL1           | -              | 2 YR  | LUBE OIL SYSTEM                                                    | Replace oil filter at every engine oil change, or when the time limit (years) is reached, at the latest                                                                                    | X      | WM00067 |
| QL1           | -              | 2 YR  | COOLANT SYSTEM                                                     | Replace filter                                                                                                                                                                             |        | WM00116 |
| QL1           | -              | 5 YR  | ENGINE MOUNTING                                                    | Measure height of rubber element                                                                                                                                                           |        | WM00126 |
| QL1           | -              | 5 YR  | ENGINE MOUNTING                                                    | Measure buffer play, adjust if necessary                                                                                                                                                   |        | WM00128 |
| QL1           | 500            | 2 YR  | LUBE OIL SYSTEM                                                    | Automatic centrifugal oil filter, check layer thickness of oil residue, clean, and replace paper element at every engine oil change at the latest                                          |        | WM00063 |
| QL1           | 500            | 9 YR  | AUTOMATIC OIL FILTER                                               | Automatic oil filter, check and clean filtering screen                                                                                                                                     | X      | WM00066 |
| QL1           | 1000           | 1 YR  | AIR SUPPLY SYSTEM                                                  | Clean air filter, empty dust collector box                                                                                                                                                 | X      | WM00149 |
| QL1           | 1000           | 2 YR  | FUEL SYSTEM (LOW PRESSURE)                                         | Fuel prefilter, replace filter                                                                                                                                                             | X      | WM00084 |
| QL1           | 1000           | 2 YR  | FUEL SYSTEM (LOW PRESSURE)                                         | Replace easy-change fuel filter                                                                                                                                                            |        | WM00086 |
| QL1           | 1000           | 2 YR  | AUXILIARY POWER TAKE-OFF                                           | Check belt condition, replace if necessary. Adjust tension.                                                                                                                                | X      | WM00170 |
| QL1           | 3000           | 1 YR  | ENGINE MOUNTING                                                    | Check general condition of rubber mounts (visual inspection)                                                                                                                               |        | WM00050 |
| QL1           | 3000           | 1 YR  | ENGINE MOUNTING                                                    | Check securing screws for secure seating                                                                                                                                                   |        | WM00136 |
| QL1           | 3000           | 3 YR  | AIR SUPPLY SYSTEM                                                  | Replace air filter                                                                                                                                                                         | X      | WM00076 |
| QL1           | 3000           | 3 YR  | COOLANT SYSTEM                                                     | Replace valve cover                                                                                                                                                                        | X      | WM00664 |
| QL1           | 3000           | 5 YR  | CRANKCASE BREATHER                                                 | Replace filter or filter element (HE200)                                                                                                                                                   | X      | WM00692 |
| QL1           | 3000           | 18 YR | BATTERY-CHARGING GENERATOR                                         | Check condition of coupling                                                                                                                                                                | X      | WM00052 |
| QL1           | 3000           | 18 YR | VALVE GEAR                                                         | Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul. |        | WM00094 |
| QL1           | 3000           | 18 YR | GROUNDING SYSTEM                                                   | Replace carbon brushes at flywheel                                                                                                                                                         | X      | WM00961 |
| QL1           | 3000           | 18 YR | GROUNDING SYSTEM                                                   | Replace carbon brushes and cylindrical roller bearings of grounding equipment at vibration damper                                                                                          | X      | WM00962 |
| QL1           | 4500           | 3 YR  | CRANKCASE BREATHER                                                 | Replace filter or filter element                                                                                                                                                           | X      | WM00119 |
| QL1           | 6750           | 9 YR  | RUNNING GEAR                                                       | Inspect cylinder liner with endoscope                                                                                                                                                      |        | WM00110 |
| QL1           | 6750           | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace fuel injection valve/injector                                                                                                                                                      |        | WM00090 |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

| Qualification | Interval [HRS] | Limit | Item                                                               | Maintenance tasks                                                                                                          | Option | Task    |
|---------------|----------------|-------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------|---------|
| QL1           | 6750           | 18 YR | CYLINDER HEAD                                                      | Measure valve protrusion                                                                                                   |        | WM00098 |
| QL3           | 13500          | 6 YR  | RUBBER SLEEVE                                                      | Replace rubber sleeve                                                                                                      |        | WM00008 |
| QL3           | 13500          | 9 YR  | AUTOMATIC OIL FILTER                                               | Automatic oil filter, remove and clean filter housing, replace filter element and bearing                                  | X      | WM00065 |
| QL3           | 13500          | 18 YR | COOLANT SYSTEM                                                     | Check LT thermostat housing and replace thermal actuator                                                                   |        | WM00054 |
| QL3           | 13500          | 18 YR | COOLANT SYSTEM                                                     | Check HT thermostat housing and replace thermal actuator                                                                   |        | WM00056 |
| QL3           | 13500          | 18 YR | LUBE OIL SYSTEM                                                    | Check centrifugal oil filter for wear                                                                                      |        | WM00062 |
| QL3           | 13500          | 18 YR | VALVE GEAR                                                         | Check rocker arm and valve bridge for wear. Visually inspect swing follower with endoscope through pushrod bore            |        | WM00093 |
| QL3           | 13500          | 18 YR | VALVE GEAR                                                         | Visually inspect camshaft running surfaces with endoscope through pushrod bore                                             |        | WM00096 |
| QL3           | 13500          | 18 YR | AUXILIARY POWER TAKE-OFF                                           | Replace auxiliary power take-off                                                                                           | X      | WM00283 |
| QL3           | 13500          | 18 YR | COOLANT SYSTEM                                                     | Replace sealing material for installation of coolant pump(s) and thermostat(s) (only LT and HT)                            |        | WM00921 |
| QL3           | 13500          | 18 YR | EXHAUST TURBOCHARGER                                               | Overhaul or replace exhaust turbocharger                                                                                   |        | WM00970 |
| QL3           | 13500          | 18 YR | STARTING SYSTEM                                                    | Overhaul or replace starter                                                                                                |        | WM00971 |
| QL3           | 13500          | 18 YR | COOLANT SYSTEM                                                     | Overhaul or replace coolant pump in LT circuit                                                                             |        | WM00973 |
| QL3           | 13500          | 18 YR | COOLANT SYSTEM                                                     | Overhaul or replace coolant pump in HT circuit                                                                             |        | WM00974 |
| QL3           | 13500          | 18 YR | BATTERY-CHARGING GENERATOR                                         | Overhaul or replace battery-charging generator                                                                             | X      | WM00976 |
| QL3           | 13500          | 18 YR | LUBE OIL SYSTEM                                                    | Clean oil cooler and check for leaks or replace                                                                            |        | WM00981 |
| QL3           | 27000          | 6 YR  | HOSE                                                               | Replace hose line                                                                                                          |        | WM00006 |
| QL3           | 27000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace HP fuel sensor                                                                                                     |        | WM00011 |
| QL3           | 27000          | 18 YR | EXHAUST SYSTEM                                                     | Replace exhaust pipe bellows                                                                                               |        | WM00072 |
| QL3           | 27000          | 18 YR | FUEL SYSTEM (LOW PRESSURE)                                         | Replace pump                                                                                                               |        | WM00087 |
| QL3           | 27000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace HP pump                                                                                                            |        | WM00091 |
| QL3           | 27000          | 18 YR | CYLINDER HEAD                                                      | Overhaul or replace cylinder head                                                                                          |        | WM00980 |
| QL4           | -              | 15 YR | STARTING SYSTEM                                                    | Replace CAPOS                                                                                                              | X      | WM00009 |
| QL4           | 27000          | 18 YR | ENGINE, GENERAL                                                    | Replace elastomer parts and seals                                                                                          |        | WM00001 |
| QL4           | 27000          | 18 YR | ENGINE, GENERAL                                                    | Disassemble engine completely. Check engine components according to assembly instructions, replace or repair as necessary. |        | WM00002 |
| QL4           | 27000          | 18 YR | BENDING / TORSIONALLY RESILIENT COUPLING AND STEEL-SPRING COUPLING | Check coupling, replace if necessary                                                                                       | X      | WM00045 |
| QL4           | 27000          | 18 YR | OIL REPLENISHING UNIT                                              | Replace gear pump                                                                                                          | X      | WM00061 |
| QL4           | 27000          | 18 YR | LUBE OIL SYSTEM                                                    | Check pump, replace if necessary                                                                                           |        | WM00070 |
| QL4           | 27000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace pressure relief valve                                                                                              |        | WM00088 |
| QL4           | 27000          | 18 YR | VALVE GEAR                                                         | Replace swing follower and swing follower shaft                                                                            |        | WM00092 |
| QL4           | 27000          | 18 YR | VALVE GEAR                                                         | Replace thrust bearing flange                                                                                              |        | WM00095 |
| QL4           | 27000          | 18 YR | RUNNING GEAR                                                       | Replace conrod bearings                                                                                                    |        | WM00099 |
| QL4           | 27000          | 18 YR | RUNNING GEAR                                                       | Replace rings                                                                                                              |        | WM00100 |
| QL4           | 27000          | 18 YR | RUNNING GEAR                                                       | Replace crankshaft bearings                                                                                                |        | WM00103 |
| QL4           | 27000          | 18 YR | GEAR TRAIN                                                         | Check tooth flanks in gear train for wear (visual inspection), replace bushing                                             |        | WM00104 |
| QL4           | 27000          | 18 YR | RUNNING GEAR                                                       | Replace cylinder liners                                                                                                    |        | WM00108 |
| QL4           | 27000          | 18 YR | CAMSHAFT                                                           | Replace camshaft bearings                                                                                                  |        | WM00111 |
| QL4           | 27000          | 18 YR | ENGINE MOUNTING                                                    | Replace rubber element of rubber mounts                                                                                    |        | WM00127 |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

TIM-ID: 0000000000 - 000

| Qualification                                           | Interval<br>[HRS] | Limit | Item                                     | Maintenance tasks                                                | Option | Task    |
|---------------------------------------------------------|-------------------|-------|------------------------------------------|------------------------------------------------------------------|--------|---------|
| QL4                                                     | 27000             | 18 YR | SENSOR TECHNOLOGY AND WIRING HARNESS     | Replace wiring harness                                           |        | WM00137 |
| QL4                                                     | 27000             | 18 YR | GROUNDING SYSTEM                         | Check flywheel (insulated), overhaul if necessary                | X      | WM00139 |
| QL4                                                     | 27000             | 18 YR | RUNNING GEAR                             | Replace conrod                                                   |        | WM00353 |
| QL4                                                     | 27000             | 18 YR | RUNNING GEAR                             | Replace vibration damper                                         |        | WM00439 |
| QL4                                                     | 27000             | 18 YR | BENDING / TORSIONALLY RESILIENT COUPLING | Replace coupling or coupling element(s)                          | X      | WM00925 |
| QL4                                                     | 27000             | 18 YR | VALVE GEAR                               | Check rocker arm and valve bridge for wear, replace as necessary |        | WM00936 |
| QL4                                                     | 27000             | 18 YR | EXHAUST SYSTEM                           | Replace exhaust pipes                                            |        | WM01030 |
| DAY = Days<br>MON = Months<br>YR = Years<br>HRS = Hours |                   |       |                                          |                                                                  |        |         |

## 1.12 Maintenance Schedule Matrix 24000 HRS

### 500 to 15000 Operating hours

| Task    | Limit | Operating hours [HRS] |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 500                   | 1000 | 1500 | 2000 | 2500 | 3000 | 3500 | 4000 | 4500 | 5000 | 5500 | 6000 | 6500 | 7000 | 7500 | 8000 | 8500 | 9000 | 9500 | 10000 | 10500 | 11000 | 11500 | 12000 | 12500 | 13000 | 13500 | 14000 | 14500 | 15000 |
| WM00285 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00286 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00287 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00288 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00289 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00290 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00291 | 1 DAY |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00881 | 1 MON |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00067 | 2 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00116 | 2 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00126 | 5 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00128 | 5 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00063 | 2 YR  | X                     | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00066 | 9 YR  | X                     | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X    | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |
| WM00149 | 1 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00084 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00086 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00170 | 2 YR  |                       | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X    |      | X     |       | X     |       | X     |       | X     |       | X     |       | X     |
| WM00050 | 1 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00136 | 1 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00076 | 3 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00664 | 3 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00692 | 5 YR  |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00052 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00094 | 18 YR |                       | X    |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00961 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00962 | 18 YR |                       |      |      |      |      | X    |      |      |      |      |      | X    |      |      |      |      |      | X    |      |       |       |       | X     |       |       |       |       |       |       | X     |
| WM00119 | 3 YR  |                       |      |      |      |      |      |      |      |      |      |      | X    |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00110 | 9 YR  |                       |      |      |      |      |      |      |      |      |      |      | X    |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00090 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      | X    |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00098 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      | X    |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00008 | 6 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00054 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00056 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00062 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00093 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00096 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00283 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00921 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00970 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |
| WM00971 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       | X     |       |       |       |       |       |       |       |

DAY = Days

MON = Months

YR = Years

HRS = Hours

TIM-ID: 0000000000 - 000

| Task    | Limit | Operating hours [HRS] |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
|---------|-------|-----------------------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|
|         |       | 500                   | 1000 | 1500 | 2000 | 2500 | 3000 | 3500 | 4000 | 4500 | 5000 | 5500 | 6000 | 6500 | 7000 | 7500 | 8000 | 8500 | 9000 | 9500 | 10000 | 10500 | 11000 | 11500 | 12000 | 12500 | 13000 | 13500 | 14000 | 14500 | 15000 |
| WM00973 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       | X     |       |       |       |       |       |       |
| WM00974 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       | X     |       |       |       |       |       |       |
| WM00976 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       | X     |       |       |       |       |       |       |
| WM00981 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       | X     |       |       |       |       |       |       |
| WM00006 | 6 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00065 | 9 YR  |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00011 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00072 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00087 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00091 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00980 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00009 | 15 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00001 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00002 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00045 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00061 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00070 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00088 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00092 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00095 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00099 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00100 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00103 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00104 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00108 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00111 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00127 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00137 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00139 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00353 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00439 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00925 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM00936 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |
| WM01030 | 18 YR |                       |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |      |       |       |       |       |       |       |       |       |       |       |       |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

## 15500 to 24000 Operating hours

| Task    | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |
|---------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|--|--|--|--|--|--|--|--|--|--|
|         |       | 15500                 | 16000 | 16500 | 17000 | 17500 | 18000 | 18500 | 19000 | 19500 | 20000 | 20500 | 21000 | 21500 | 22000 | 22500 | 23000 | 23500 | 24000 |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00285 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |
| WM00286 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |  |  |  |  |  |  |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

| Task    | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
|---------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|--|--|--|--|
|         |       | 15500                 | 16000 | 16500 | 17000 | 17500 | 18000 | 18500 | 19000 | 19500 | 20000 | 20500 | 21000 | 21500 | 22000 | 22500 | 23000 | 23500 | 24000 |  |  |  |  |  |  |
| WM00287 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00288 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00289 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00290 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00291 | 1 DAY |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00881 | 1 MON |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00067 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00116 | 2 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00126 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00128 | 5 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |  |  |  |  |
| WM00063 | 2 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |  |  |  |  |  |  |
| WM00066 | 9 YR  | X                     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     | X     |  |  |  |  |  |  |
| WM00149 | 1 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |  |  |  |  |  |  |
| WM00084 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |  |  |  |  |  |  |
| WM00086 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |  |  |  |  |  |  |
| WM00170 | 2 YR  |                       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |       | X     |  |  |  |  |  |  |
| WM00050 | 1 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00136 | 1 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00076 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00664 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00692 | 5 YR  |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00052 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00094 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00961 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00962 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       | X     |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00119 | 3 YR  |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00110 | 9 YR  |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00090 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00098 | 18 YR |                       |       |       |       |       | X     |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00008 | 6 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00054 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00056 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00062 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00093 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00096 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00283 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00921 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00970 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00971 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00973 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00974 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00976 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00981 | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00006 | 6 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |
| WM00065 | 9 YR  |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |  |  |  |  |

DAY = Days  
MON = Months  
YR = Years  
HRS = Hours

| Task         | Limit | Operating hours [HRS] |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
|--------------|-------|-----------------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|              |       | 15500                 | 16000 | 16500 | 17000 | 17500 | 18000 | 18500 | 19000 | 19500 | 20000 | 20500 | 21000 | 21500 | 22000 | 22500 | 23000 | 23500 | 24000 |  |  |
| WM00011      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00072      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00087      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00091      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00980      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00009      | 15 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| WM00001      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00002      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00045      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00061      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00070      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00088      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00092      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00095      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00099      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00100      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00103      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00104      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00108      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00111      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00127      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00137      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00139      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00353      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00439      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00925      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM00936      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| WM01030      | 18 YR |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       | X     |  |  |
| DAY = Days   |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| MON = Months |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| YR = Years   |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |
| HRS = Hours  |       |                       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |       |  |  |

## 1.13 Cyclical maintenance tasks 24000 HRS

| Qualification | Interval [HRS] | Limit | Item                                                               | Maintenance tasks                                                                                                                                                                          | Option | Task    |
|---------------|----------------|-------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|---------|
| <b>Engine</b> |                |       |                                                                    |                                                                                                                                                                                            |        |         |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check engine oil level                                                                                                                                                                     |        | WM00285 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Visually inspect for leaks and general condition of engine                                                                                                                                 |        | WM00286 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check maintenance indicator of air filter                                                                                                                                                  | X      | WM00287 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check relief bore of coolant pump                                                                                                                                                          |        | WM00288 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | When engine is running, check for abnormal running noises, exhaust gas color and vibrations                                                                                                |        | WM00289 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Drain water and contaminants from fuel prefilter                                                                                                                                           | X      | WM00290 |
| QL 1          | -              | 1 DAY | ENGINE OPERATIONAL MONITORING                                      | Check pointer of vacuum gauge of fuel prefilter                                                                                                                                            | X      | WM00291 |
| QL 1          | -              | 1 MON | BENDING / TORSIONALLY RESILIENT COUPLING AND STEEL-SPRING COUPLING | Check coupling (visual inspection), replace as necessary                                                                                                                                   | X      | WM00881 |
| QL 1          | -              | 2 YR  | LUBE OIL SYSTEM                                                    | Replace oil filter at every engine oil change, or when the time limit (years) is reached, at the latest                                                                                    | X      | WM00067 |
| QL 1          | -              | 2 YR  | COOLANT SYSTEM                                                     | Replace filter                                                                                                                                                                             |        | WM00116 |
| QL 1          | -              | 5 YR  | ENGINE MOUNTING                                                    | Measure height of rubber element                                                                                                                                                           |        | WM00126 |
| QL 1          | -              | 5 YR  | ENGINE MOUNTING                                                    | Measure buffer play, adjust if necessary                                                                                                                                                   |        | WM00128 |
| QL 1          | 500            | 2 YR  | LUBE OIL SYSTEM                                                    | Automatic centrifugal oil filter, check layer thickness of oil residue, clean, and replace paper element at every engine oil change at the latest                                          |        | WM00063 |
| QL 1          | 500            | 9 YR  | AUTOMATIC OIL FILTER                                               | Automatic oil filter, check and clean filtering screen                                                                                                                                     | X      | WM00066 |
| QL 1          | 1000           | 1 YR  | AIR SUPPLY SYSTEM                                                  | Clean air filter, empty dust collector box                                                                                                                                                 | X      | WM00149 |
| QL 1          | 1000           | 2 YR  | FUEL SYSTEM (LOW PRESSURE)                                         | Fuel prefilter, replace filter                                                                                                                                                             | X      | WM00084 |
| QL 1          | 1000           | 2 YR  | FUEL SYSTEM (LOW PRESSURE)                                         | Replace easy-change fuel filter                                                                                                                                                            |        | WM00086 |
| QL 1          | 1000           | 2 YR  | AUXILIARY POWER TAKE-OFF                                           | Check belt condition, replace if necessary. Adjust tension.                                                                                                                                | X      | WM00170 |
| QL 1          | 3000           | 1 YR  | ENGINE MOUNTING                                                    | Check general condition of rubber mounts (visual inspection)                                                                                                                               |        | WM00050 |
| QL 1          | 3000           | 1 YR  | ENGINE MOUNTING                                                    | Check securing screws for secure seating                                                                                                                                                   |        | WM00136 |
| QL 1          | 3000           | 3 YR  | AIR SUPPLY SYSTEM                                                  | Replace air filter                                                                                                                                                                         | X      | WM00076 |
| QL 1          | 3000           | 3 YR  | COOLANT SYSTEM                                                     | Replace valve cover                                                                                                                                                                        | X      | WM00664 |
| QL 1          | 3000           | 5 YR  | CRANKCASE BREATHER                                                 | Replace filter or filter element (HE200)                                                                                                                                                   | X      | WM00692 |
| QL 1          | 3000           | 18 YR | BATTERY-CHARGING GENERATOR                                         | Check condition of coupling                                                                                                                                                                | X      | WM00052 |
| QL 1          | 3000           | 18 YR | VALVE GEAR                                                         | Check valve clearance, adjust if necessary. ATTENTION! First adjustment after 1,000 operating hours on a new engine and after 1,000 operating hours following each cylinder head overhaul. |        | WM00094 |
| QL 1          | 3000           | 18 YR | GROUNDING SYSTEM                                                   | Replace carbon brushes at flywheel                                                                                                                                                         | X      | WM00961 |
| QL 1          | 3000           | 18 YR | GROUNDING SYSTEM                                                   | Replace carbon brushes and cylindrical roller bearings of grounding equipment at vibration damper                                                                                          | X      | WM00962 |
| QL 1          | 6000           | 3 YR  | CRANKCASE BREATHER                                                 | Replace filter or filter element                                                                                                                                                           | X      | WM00119 |
| QL 1          | 6000           | 9 YR  | RUNNING GEAR                                                       | Inspect cylinder liner with endoscope                                                                                                                                                      |        | WM00110 |
| QL 1          | 6000           | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace fuel injection valve/injector                                                                                                                                                      |        | WM00090 |

DAY = Days  
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TIM-ID: 0000000000 - 000

| Qualification | Interval [HRS] | Limit | Item                                                               | Maintenance tasks                                                                                                          | Option | Task    |
|---------------|----------------|-------|--------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|--------|---------|
| QL1           | 6000           | 18 YR | CYLINDER HEAD                                                      | Measure valve protrusion                                                                                                   |        | WM00098 |
| QL3           | 12000          | 6 YR  | RUBBER SLEEVE                                                      | Replace rubber sleeve                                                                                                      |        | WM00008 |
| QL3           | 12000          | 18 YR | COOLANT SYSTEM                                                     | Check LT thermostat housing and replace thermal actuator                                                                   |        | WM00054 |
| QL3           | 12000          | 18 YR | COOLANT SYSTEM                                                     | Check HT thermostat housing and replace thermal actuator                                                                   |        | WM00056 |
| QL3           | 12000          | 18 YR | LUBE OIL SYSTEM                                                    | Check centrifugal oil filter for wear                                                                                      |        | WM00062 |
| QL3           | 12000          | 18 YR | VALVE GEAR                                                         | Check rocker arm and valve bridge for wear. Visually inspect swing follower with endoscope through pushrod bore            |        | WM00093 |
| QL3           | 12000          | 18 YR | VALVE GEAR                                                         | Visually inspect camshaft running surfaces with endoscope through pushrod bore                                             |        | WM00096 |
| QL3           | 12000          | 18 YR | AUXILIARY POWER TAKE-OFF                                           | Replace auxiliary power take-off                                                                                           | X      | WM00283 |
| QL3           | 12000          | 18 YR | COOLANT SYSTEM                                                     | Replace sealing material for installation of coolant pump(s) and thermostat(s) (only LT and HT)                            |        | WM00921 |
| QL3           | 12000          | 18 YR | EXHAUST TURBOCHARGER                                               | Overhaul or replace exhaust turbocharger                                                                                   |        | WM00970 |
| QL3           | 12000          | 18 YR | STARTING SYSTEM                                                    | Overhaul or replace starter                                                                                                |        | WM00971 |
| QL3           | 12000          | 18 YR | COOLANT SYSTEM                                                     | Overhaul or replace coolant pump in LT circuit                                                                             |        | WM00973 |
| QL3           | 12000          | 18 YR | COOLANT SYSTEM                                                     | Overhaul or replace coolant pump in HT circuit                                                                             |        | WM00974 |
| QL3           | 12000          | 18 YR | BATTERY-CHARGING GENERATOR                                         | Overhaul or replace battery-charging generator                                                                             | X      | WM00976 |
| QL3           | 12000          | 18 YR | LUBE OIL SYSTEM                                                    | Clean oil cooler and check for leaks or replace                                                                            |        | WM00981 |
| QL3           | 24000          | 6 YR  | HOSE                                                               | Replace hose line                                                                                                          |        | WM00006 |
| QL3           | 24000          | 9 YR  | AUTOMATIC OIL FILTER                                               | Automatic oil filter, remove and clean filter housing, replace filter element and bearing                                  | X      | WM00065 |
| QL3           | 24000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace HP fuel sensor                                                                                                     |        | WM00011 |
| QL3           | 24000          | 18 YR | EXHAUST SYSTEM                                                     | Replace exhaust pipe bellows                                                                                               |        | WM00072 |
| QL3           | 24000          | 18 YR | FUEL SYSTEM (LOW PRESSURE)                                         | Replace pump                                                                                                               |        | WM00087 |
| QL3           | 24000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace HP pump                                                                                                            |        | WM00091 |
| QL3           | 24000          | 18 YR | CYLINDER HEAD                                                      | Overhaul or replace cylinder head                                                                                          |        | WM00980 |
| QL4           | -              | 15 YR | STARTING SYSTEM                                                    | Replace CAPOS                                                                                                              | X      | WM00009 |
| QL4           | 24000          | 18 YR | ENGINE, GENERAL                                                    | Replace elastomer parts and seals                                                                                          |        | WM00001 |
| QL4           | 24000          | 18 YR | ENGINE, GENERAL                                                    | Disassemble engine completely. Check engine components according to assembly instructions, replace or repair as necessary. |        | WM00002 |
| QL4           | 24000          | 18 YR | BENDING / TORSIONALLY RESILIENT COUPLING AND STEEL-SPRING COUPLING | Check coupling, replace if necessary                                                                                       | X      | WM00045 |
| QL4           | 24000          | 18 YR | OIL REPLENISHING UNIT                                              | Replace gear pump                                                                                                          | X      | WM00061 |
| QL4           | 24000          | 18 YR | LUBE OIL SYSTEM                                                    | Check pump, replace if necessary                                                                                           |        | WM00070 |
| QL4           | 24000          | 18 YR | FUEL SYSTEM (HIGH PRESSURE)                                        | Replace pressure relief valve                                                                                              |        | WM00088 |
| QL4           | 24000          | 18 YR | VALVE GEAR                                                         | Replace swing follower and swing follower shaft                                                                            |        | WM00092 |
| QL4           | 24000          | 18 YR | VALVE GEAR                                                         | Replace thrust bearing flange                                                                                              |        | WM00095 |
| QL4           | 24000          | 18 YR | RUNNING GEAR                                                       | Replace conrod bearings                                                                                                    |        | WM00099 |
| QL4           | 24000          | 18 YR | RUNNING GEAR                                                       | Replace rings                                                                                                              |        | WM00100 |
| QL4           | 24000          | 18 YR | RUNNING GEAR                                                       | Replace crankshaft bearings                                                                                                |        | WM00103 |
| QL4           | 24000          | 18 YR | GEAR TRAIN                                                         | Check tooth flanks in gear train for wear (visual inspection), replace bushing                                             |        | WM00104 |
| QL4           | 24000          | 18 YR | RUNNING GEAR                                                       | Replace cylinder liners                                                                                                    |        | WM00108 |
| QL4           | 24000          | 18 YR | CAMSHAFT                                                           | Replace camshaft bearings                                                                                                  |        | WM00111 |
| QL4           | 24000          | 18 YR | ENGINE MOUNTING                                                    | Replace rubber element of rubber mounts                                                                                    |        | WM00127 |

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|---------------------------------------------------------|-------------------|-------|------------------------------------------|------------------------------------------------------------------|--------|---------|
| QL4                                                     | 24000             | 18 YR | SENSOR TECHNOLOGY AND WIRING HARNESS     | Replace wiring harness                                           |        | WM00137 |
| QL4                                                     | 24000             | 18 YR | GROUNDING SYSTEM                         | Check flywheel (insulated), overhaul if necessary                | X      | WM00139 |
| QL4                                                     | 24000             | 18 YR | RUNNING GEAR                             | Replace conrod                                                   |        | WM00353 |
| QL4                                                     | 24000             | 18 YR | RUNNING GEAR                             | Replace vibration damper                                         |        | WM00439 |
| QL4                                                     | 24000             | 18 YR | BENDING / TORSIONALLY RESILIENT COUPLING | Replace coupling or coupling element(s)                          | X      | WM00925 |
| QL4                                                     | 24000             | 18 YR | VALVE GEAR                               | Check rocker arm and valve bridge for wear, replace as necessary |        | WM00936 |
| QL4                                                     | 24000             | 18 YR | EXHAUST SYSTEM                           | Replace exhaust pipes                                            |        | WM01030 |
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